SIDDAGANGA INSTITUTE OF TECHNOLOGY, TUMAKURU-572103

(An Autonomous Institute under Visvesvaraya Technological University, Belagavi)

Major Project Report (Phase-III) on

"EEG Based Cognitive Assessment Model"

submitted in partial fulfillment of the completion of VII semester of

BACHELOR OF ENGINEERING

in

ELECTRONICS & COMMUNICATION ENGINEERING

ELECTRONICS & TELECOMMUNICATION ENGINEERING

Submitted by

Poorvi	1SI23EC079
Pratham Srivastava	1SI23EC081
Devendra M	1SI23ET011
Gowri S	1SI23ET015

under the guidance of

Dr. Veena Karjigi
Associate Professor
Department of E&CE
SIT, Tumakuru

Dr. Chandrashekar H M
Assistant Professor
Department of E&TE
SIT, Tumakuru

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING &
DEPARTMENT OF ELECTRONICS & TELECOMMUNICATION ENGINEERING

2026-27

SIDDAGANGA INSTITUTE OF TECHNOLOGY, TUMAKURU-572103

(An Autonomous Institute under Visvesvaraya Technological University, Belagavi)

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

CERTIFICATE

Certified that the project work entitled "EEG BASED COGNITIVE ASSESSMENT MODEL" is a bonafide work carried out by Poorvi (1SI23EC079), Pratham Srivastava (1SI23EC081), Devendra M (1SI23ET011) and Gowri S (1SI23ET015) in partial fulfillment for the completion of VII Semester of Bachelor of Engineering in Electronics & Communication Engineering and Electronics & Telecommunication Engineering from Siddaganga Institute of Technology, an autonomous institute under Visvesvaraya Technological University, Belagavi, during the academic year 2026–27. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the department library. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the completion of VII semester of Bachelor of Engineering.

Dr. Veena Karjigi
Associate Professor
Dept. of ECE
SIT, Tumakuru-03

Head of the Department
Dept. of ECE
SIT, Tumakuru-03

Dr. Chandrashekar H M
Assistant Professor
Dept. of ETE
SIT, Tumakuru-03

Head of the Department
Dept. of ETE
SIT, Tumakuru-03

External Viva:

Names of the Examiners

1.

2.

Signature with date

ACKNOWLEDGEMENT

We offer our humble pranams at the lotus feet of **His** Holiness, **Dr. Sree Sree Sivakumara Swamigalu**, Founder President and **His** Holiness, **Sree Sree Siddalinga Swamigalu**, President, Sree Siddaganga Education Society, Sree Siddaganga Math for bestowing upon their blessings.

We deem it as a privilege to thank **Dr. Shivakumaraiah**, CEO, SIT, Tumakuru, and **Dr. S V Dinesh**, Principal, SIT, Tumakuru for fostering an excellent academic environment in this institution, which made this endeavor fruitful.

We would like to express our sincere gratitude to **Dr. R Kumarswamy**, Professor and Head, Department of E&CE, SIT, Tumakuru for his encouragement and valuable suggestions.

We would like to express our sincere gratitude to **Dr. K Vishwanath**, Professor and Head, Department of E&TE, SIT, Tumakuru for his encouragement and valuable suggestions.

We thank our guides **Dr. Veena Karjigi**, Associate Professor, Department of Electronics & Communication Engineering and **Dr. Chandrashekar H M**, Assistant Professor, Department of Electronics & Telecommunication Engineering SIT, Tumakuru for the valuable guidance, advice and encouragement.

Poorvi	(1SI23EC079)
Pratham Srivastava	(1SI23EC081)
Devendra M	(1SI23ET011)
Gowri S	(1SI23ET015)

Course Outcomes

After successful completion of major project, graduates will be able to:

CO1: To identify a problem through literature survey and knowledge of contemporary engineering technology

CO2: To consolidate the literature search to identify issues/gaps and formulate the engineering problem

CO3: To prepare project schedule for the identified design methodology and engage in budget analysis, and share responsibility for every member in the team

CO4: To provide sustainable engineering solution considering health, safety, legal, cultural issues and also demonstrate concern for environment

CO5: To identify and apply the mathematical concepts, science concepts, engineering and management concepts necessary to implement the identified engineering problem

CO6: To select the engineering tools/components required to implement the proposed solution for the identified engineering problem

CO7: To analyze, design, and implement optimal design solution, interpret results of experiments and draw valid conclusion

CO8: To demonstrate effective written communication through the project report, the one-page poster presentation, and preparation of the video about the project and the four page IEEE/Springer paper format of the work

CO9: To engage in effective oral communication through power point presentation and demonstration of the project work

CO10: To demonstrate compliance to the prescribed standards/safety norms and abide by the norms of professional ethics

CO11: To perform in the team, contribute to the team and mentor/lead the team

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO-1											3		3
CO-2		3										3	
CO-3										3			3
CO-4						3							3
CO-5	3	3										3	
CO-6					3						3		3
CO-7			3	3								3	
CO-8										3			3
CO-9										3			3
CO-10								3					3
CO-11									3				3
Average	3	3	3	3	3	3	3	3	3	3	3	3	3

Attainment level: - 1: Slight (low) 2: Moderate (medium) 3: Substantial (high)

Knowledge and Attitude Profile (WK)

WK1: A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.

WK2: Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.

WK3: A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.

WK4: Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.

WK5: Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.

WK6: Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.

WK7: Knowledge of the role of engineering in society and identified issues in engineering

practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.

WK8: Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

WK9: Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

Program Outcomes (POs)

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop the solution of complex engineering problems

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for public health and safety, whole-life cost, net zero carbon, culture, society and environment as required (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data to provide valid conclusions (WK8)

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7)

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national and international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

Program Specific Outcomes (PSOs)

PSO1: The ability to analyze and design systems in the areas related to microelectronics, communication, signal processing and embedded systems for solving real world problems (Professional Skills).

PSO2: The ability to identify problems in the areas of communication and embedded systems and provide efficient solutions using modern tools/algorithm individually or working in a team (Problem Solving Skills).

Abstract

The brain keeps producing small electrical signals all the time, and these signals change with what a person is doing and how their mind is working. This project uses a wearable, four-channel dry-electrode EEG headset to read two kinds of information from the same recording. The first is which hand a person is imagining moving, left or right, without any actual movement. This is called motor imagery and it is a basic building block of a brain-computer interface. The second is the mental state of the person, given as a cognitive score built from the power in the theta, alpha and beta frequency bands, which covers attention, cognitive load, relaxation, stress and fatigue.

For motor imagery, the analysis method was first checked on a public dataset of 50 acute stroke patients, and then on the public BCI Competition IV 2b dataset of nine healthy subjects, which uses the same 250 Hz sampling rate and the same C3 and C4 electrode positions as the headset. On the 2b data the expected drop in 8–30 Hz power over the side of the head opposite to the imagined hand was seen in 16 of 18 cases, and the ratio of C3 to C4 power was higher for the left hand than for the right hand in all nine subjects. Used alone, this ratio identified the hand on unseen days with 73.5% accuracy, and the best SVM-based classifier reached 80.2% on the same person and 74.7% on people it had never seen. Left and right motor imagery was also recorded with the headset, where the same C3/C4 difference appears in the trials examined, but classifiers trained on either public dataset stayed near chance on the recorded data. This shows that the recording protocol, rather than the classifier, is the main thing to improve. The cognitive scoring part works end to end, with the scores computed live against each person's own calibration. No audio is recorded at any stage.

Contents

Abstract **i**

List of Figures **iii**

List of Tables **iv**

1 Introduction **1**

1.1 Motivation 2

1.2 Objective of the Project 3

1.3 Organization of the Report 3

2 Literature Survey **5**

2.1 Wearable EEG Systems 5

2.2 Motor Imagery EEG Datasets 6

2.3 Feature Extraction and Classification for Motor Imagery 7

2.4 Decoding Speech from EEG 9

2.5 EEG for Cognitive and Mental State Assessment 9

2.6 Summary of the Literature Survey 10

3 System Overview **12**

4 System Hardware **15**

4.1 EEG Recorder 15

4.1.1 Wet Electrode EEG System 15

4.1.2 Dry Electrode EEG System 15

4.1.3 GELO BCI Headset 16

4.2 Raspberry Pi 5 18

4.3 Aluminum Heat Sink Case with Dual Fans for Raspberry Pi 5 19

4.3.1 Features of Aluminum Heat Sink 19

4.4 Waveshare 7-Inch LCD Display 20

4.4.1 Features of Waveshare LCD Display 20

5 System Software **22**

5.1 Software Requirements 22

5.2 Flowchart 22

6 Results **26**

6.1 Evaluation Aspects 26

6.1.1 Cases of Comparison 26

6.1.2 Parameters Under Study 27

6.2 Experimental Setup 27

6.3 Reproduction of the Stroke Dataset Analysis 28

6.4 Results on the BCI Competition IV 2b Dataset 29

6.4.1 How the Dataset Was Used 30

6.4.2 ERD/ERS of the BCI Competition IV 2b Subjects 32

6.4.3 Power Spectral Density of Subject B04 36

6.4.4 The C3/C4 Power Ratio as a Left and Right Detector 38

6.4.5 SVM Training and Classification Accuracy 40

6.4.6 Testing a BCI Competition IV 2b Model on the Recorded Data . . 43

6.5 Results on the Recorded Data 44

6.5.1 ERD/ERS of the Recorded Data 44

6.5.2 Power Spectral Density for Left and Right Imagery 45

6.5.3 Classification Results 47

6.6 Cognitive Assessment Results 48

6.6.1 Signal Quality and Filtering 49

6.6.2 Usable Frequency Range 50

6.6.3 Stress and Fatigue Reference Levels 50

6.6.4 Complete Session Output 51

6.7 Reason for the Observed Variations 52

6.8 Summary of Analysis 53

6.9 Work carried out so far 55

7 Conclusion and Future Work **56**

7.1 Conclusion 56

7.2 Future Work 57

Bibliography **59**

Appendices **63**

A Program Outcomes and Program Specific Outcomes addressed **64**

B Sustainable Development Goals addressed **67**

List of Figures

3.1 EEG Based Cognitive Assessment Overview 12

4.1 GELO BCI Headset 17
4.2 Raspberry Pi 5 19
4.3 Aluminum Heat Sink Case with Dual Fans for Raspberry Pi 5 20
4.4 Waveshare 7-Inch LCD Display 21

5.1 Flowchart of EEG based Cognitive Assessment Model 24

6.1 EEG signal acquisition using the Capsule GUI 28
6.2 ERD/ERS of C3 and C4 for subject 8 of the reference dataset 29
6.3 Timing of one trial in the BCI Competition IV 2b dataset 30
6.4 The two ways the classifiers were tested on the BCI Competition IV 2b data 31
6.5 Processing chain used for the BCI Competition IV 2b data 32
6.6 ERD/ERS of C3 and C4 for subject B04 of the BCI Competition IV 2b dataset 33
6.7 ERD/ERS of C3 and C4 for all nine subjects of the BCI Competition IV 2b dataset 35
6.8 Power spectral density of C3 and C4 for imagined left-hand movement, subject B04 of the BCI Competition IV 2b dataset 36
6.9 Power spectral density of C3 and C4 for imagined right-hand movement, subject B04 of the BCI Competition IV 2b dataset 37
6.10 C3/C4 power ratio for imagined left and right hand in each BCI Competition IV 2b subject 39
6.11 Single-trial accuracy of one-number features on the BCI Competition IV 2b dataset 40
6.12 Accuracy for each BCI Competition IV 2b subject with the best model for each test and with the C3/C4 ratio alone 42
6.13 ERD/ERS of C3 and C4 for the data recorded in this project, 8–30 Hz, $F_s = 250$ Hz 44

6.14 Power spectral density of C3 and C4 for an imagined left-hand grasp . . . 46
6.15 Power spectral density of C3 and C4 for an imagined right-hand grasp . . . 46
6.16 Electrode resistance of the four channels after connecting the headset . . . 49
6.17 Frequency response of the 0.5 Hz high-pass and 50 Hz notch filters 50
6.18 Fraction of the original amplitude retained across frequency bands 50
6.19 Arousal ratio over a session, with the rest and work reference levels marked 51
6.20 Cognitive score interface showing the session output 52

List of Tables

4.1 Main specifications of the GELO BCI headset [25] 17
4.2 Main specifications of the Raspberry Pi 5 relevant to this project [26] . . . 18

5.1 Software set-up used in this project 23
5.2 Python libraries and what each one does in this project 23

6.1 Mean ERD/ERS (%) from 0.5 to 3.5 s after the cue for every BCI Competition IV 2b subject 34
6.2 C3/C4 power ratio (8–30 Hz, 0–4 s after the cue) and single-trial accuracy of the ratio for each BCI Competition IV 2b subject 38
6.3 Accuracy of the ERD% SVM on sessions 4 and 5 of the BCI Competition IV 2b dataset, mean of nine subjects 41
6.4 Classification accuracy on the BCI Competition IV 2b dataset, mean of nine subjects 42
6.5 Confusion matrix for the new-person test using only C3 and C4, all 6520 trials (CSP + SVM with Euclidean alignment) 43
6.6 Classification accuracy of the trained SVM models on the recorded data . . 48

Chapter 1

Introduction

The human brain is one of the most complex things that exists in nature. It controls everything a person does, from walking and talking to thinking and feeling emotions. Even when a person is sitting quietly and doing nothing, the brain is still working. It never fully stops. With the growth of technology, it has become possible to study the brain in ways that were not possible before. This project is one such attempt to use modern technology to understand two important things about the brain, which are the mental state of a person and which hand a person is imagining moving.

In daily life, every person goes through different mental conditions. There are times when a person feels very fresh and focused. There are also times when a person feels mentally tired, dull, or distracted. These changes happen because of many reasons like lack of sleep, long hours of studying or working, stress, or even boredom. These different mental conditions are referred to as cognitive states. The word cognitive simply means anything related to thinking, attention, memory, or mental processing. So cognitive assessment means measuring or evaluating how the brain is functioning at a given moment. Cognitive assessment is not only useful in medical cases. It is also useful in everyday situations. For example, if a student is studying for a long time, it is helpful to know if the student is still mentally active or has become too tired to absorb new information. Similarly, in workplaces where tasks require high concentration, like air traffic control or surgery, knowing the cognitive state of the person doing the task can help avoid mistakes.

The brain produces electrical signals continuously. These signals change depending on what the brain is doing at a given moment. When a person is relaxed, the signals look different compared to when the person is focused or stressed. These signals can be measured from outside the head without any surgery or internal device. This is why brain signal based cognitive assessment has gained a lot of interest in recent years. The method used to record brain electrical signals from the scalp is called Electroencephalography, which is commonly known as EEG. In this method, small sensors called electrodes are placed on different positions on the scalp. These electrodes pick up the tiny electrical

signals produced by the brain. Along with cognitive assessment, this project looks at motor imagery. When a person only imagines moving the left or the right hand, without moving at all, the part of the brain that controls that hand still changes its activity, and this change can be picked up from electrodes placed over the motor cortex. If a computer can tell which hand was imagined, the person can control something, such as a cursor or a wheelchair, by thought alone. A longer-term aim of the work is imagined speech, where a person says a word only in their mind, and the motor imagery work is the first step towards it.

1.1 Motivation

The brain produces electrical signals all the time but reading those signals and making useful sense out of them is not easy. EEG signals are noisy, complex, and they change a lot from person to person. Getting reliable information from them, especially when using a simple wearable headset in everyday conditions, is still a difficult problem. This problem becomes even more important to solve when looking at the current global situation. The World Health Organization estimates that over 3 billion people, more than one in three, live with a neurological condition, and that such conditions now cause more ill health and disability than any other group of diseases [1]. These numbers show that there is a real need for simple and non-invasive tools that can monitor brain activity and help people in practical ways.

Now when it comes to recording brain signals, there are two ways of doing it. The invasive way involves placing electrodes inside the skull through surgery, which gives clear signals but comes with serious risks like infection, bleeding, or permanent brain damage. Because of this, non-invasive EEG is used in this project, where electrodes sit on the outside of the scalp without any surgery. Within non-invasive EEG, gel based electrodes give better signal quality but require a trained person to apply gel on each electrode point, which is messy, slow, and not practical for everyday use. Dry electrodes on the other hand need no gel, are quick to set up, and the person can wear the headset themselves without any help. Even though they may pick up slightly more noise, they are a much better choice for a real world wearable setup like this one.

One of the things this project works on is helping people who cannot speak. There are people whose brain works completely fine but due to conditions like paralysis or nerve

damage, they have no way to express what they want to say. Saying something as basic as being in pain or needing water becomes impossible for them. The brain still produces clear electrical signals when such a person only imagines a movement or a word. In this project these signals are recorded with an EEG headset, and a machine learning model is trained to tell which hand the person was imagining moving. This is the simplest form of a brain-computer interface and the first step towards decoding imagined words.

The second thing this project works on is understanding the mental condition of a person at a given moment. In daily life, whether it is a student studying for long hours, a working professional under deadline pressure, or anyone going through a mentally tiring day, people usually do not realize when their brain has stopped working well. They keep going even after their focus has already dropped. By analyzing EEG signals and calculating a cognitive score, this project offers a simple way to know the mental state of a person

1.2 Objective of the Project

The main objective of this project is to develop an EEG-based cognitive assessment system that can record and analyse brain signals for different applications. The project mainly focuses on telling left-hand from right-hand motor imagery using EEG data, and on analysing cognitive state using frequency band analysis.

- To develop an EEG database for motor imagery (MI) actuation.
- To design a motor imagery classifier system.
- To compute a cognitive score using EEG frequency band analysis.

1.3 Organization of the Report

This report is divided into seven chapters that walk through the development of the EEG-based cognitive assessment model. The first chapter explains what EEG is, why it is useful in brain-computer interfaces, and what motivated this project given the growing number of people affected by neurological disorders. The second chapter looks at existing research on wearable EEG systems, public motor imagery datasets, including the stroke patient dataset and the BCI Competition IV 2b dataset, methods for feature extraction and classification, speech decoding and mental state assessment, and ends by identifying

the gaps this project tries to address. The third chapter gives a broad overview of the entire system, from recording and cleaning EEG signals all the way to extracting features and using a machine learning model to generate an output. The fourth chapter covers the hardware, which is the GELO BCI headset, Raspberry Pi 5, aluminium heat sink casing and Waveshare LCD screen, and explains the role each component plays. The fifth chapter shifts to the software, detailing the tools and libraries used along with the step-by-step process for signal acquisition, preprocessing, feature extraction, classification and cognitive score calculation. The sixth chapter presents the results: the reproduction of the stroke dataset analysis, the ERD/ERS, power spectral density, C3/C4 power ratio and SVM results on the BCI Competition IV 2b dataset, the same analysis on the data recorded with the headset, and the cognitive assessment results. The seventh chapter summarises what was achieved and sets out the next steps, such as a better recording protocol, per-user calibration and deployment on the Raspberry Pi.

Chapter 2

Literature Survey

This chapter looks at the work that other researchers have already done in the areas this project touches. It covers wearable EEG devices, publicly available motor imagery datasets, the methods used to extract features and classify EEG signals, and the use of EEG to measure a person's mental and cognitive state. A short section on decoding speech from EEG is also included, since that is the direction this work is expected to move in later. The aim of this chapter is to understand what has already been done, what problems are still open, and where this project fits in.

2.1 Wearable EEG Systems

EEG devices have come a long way from the large machines used in hospitals. Today there are small, portable EEG headsets that a person can wear while going about normal activities. This matters for any brain-computer interface (BCI) that is meant to be used outside a lab.

Mullen and colleagues put together a complete wearable system with dry electrodes and showed that it could follow brain activity and mental state live, while the person was wearing it [2]. Because no gel is used, the headset is quick to put on and stays comfortable for a long time. Their recordings were good enough for practical use, although still a little noisier than a gel cap in a laboratory.

Kappel and colleagues went a step further and moved the dry electrodes into the ear canal, so that the recorder looks like an ordinary earphone [3]. The signals they picked up were weaker than with electrodes on the scalp, but the device was so unobtrusive that it could be worn through a normal day, which is the kind of comfort a practical brain-computer interface needs.

Both of these studies are directly relevant to the hardware used in this project. The GELO BCI headset used here is a dry-electrode headphone with only four EEG channels (A1, A2, C3 and C4) and a single reference at Cz, sampled at 250 Hz. It gains the comfort and quick setup reported in these studies, but it also inherits their limitations:

fewer channels, weaker signals and more noise than a research-grade cap. This trade-off shapes much of the work reported in the later chapters.

2.2 Motor Imagery EEG Datasets

Motor imagery (MI) means imagining a movement without actually making it. When a person imagines moving the left or right hand, the activity over the motor cortex changes, and these changes can be picked up by electrodes placed at C3 and C4. Building a classifier for this needs a good amount of labelled data, which is hard for a student project to collect from scratch.

The first reference dataset used in this project comes from Liu and colleagues, who released motor imagery recordings from people in the first month after a stroke [4]. Fifty patients took part, the youngest aged 31 and the oldest 77. Every patient did 40 trials, half imagining gripping with the left hand and half with the right, which adds up to 2000 trials. A trial lasted 8 s: 2 s of instruction, 4 s of imagery and 2 s of rest. The recordings were made at 500 Hz with a portable wireless cap using saline instead of gel, with 30 EEG electrodes, 2 EOG electrodes and a marker channel, referenced to CPz. The authors share both the raw and the cleaned data, together with clinical notes on each patient.

This dataset is important for the present work for two reasons. First, it supplies a labelled, well-documented set of left and right MI trials that can be used to train a classifier, which is the first objective of this project. Second, the paper reports its own analysis of the data, including event-related desynchronisation and a set of classification results, which gives a clear baseline to compare against. The channel layout is also convenient, as C3 and C4 are available in the dataset and are the same two positions recorded by the headset used here. The main limitation is that the subjects are stroke patients recorded with a 30-channel system, while the data collected in this project comes from healthy subjects using only four channels. This difference between the training data and the test data turns out to matter a great deal, as the results chapter shows.

A second public dataset was used to fill this gap. Dataset 2b of BCI Competition IV, which was released for the competition in 2008 and is called the BCI 2008 dataset in this report, was recorded at Graz University of Technology [5, 6]. It contains nine healthy, right-handed people who imagined moving the left or the right hand whenever an arrow pointing to that side appeared on a screen. Each person came back for five sessions on

different days. The first two sessions were plain cued imagery without feedback, about 120 trials each, and in the last three, about 160 trials each, a smiley face on the screen moved left or right according to the person's own brain signal. Only three EEG channels were recorded, over C3, Cz and C4, each as a bipolar pair of closely spaced electrodes, at 250 Hz with a 0.5–100 Hz band-pass filter and a 50 Hz notch filter. Three EOG channels were also recorded, but they are meant only for spotting eye movements and not for classification. The dataset has 6520 trials in total, half for each hand.

This dataset suits the present work for three reasons. The subjects are healthy, as in the recordings made for this project. The sampling rate is the same 250 Hz as the GELO headset, and the layout is almost the same, with the two motor cortex electrodes at C3 and C4 plus one at Cz. Finally, it comes with a fixed split, three sessions for training and two later sessions for testing, so a classifier can be checked on days it has never seen. Many methods have been tested on it, and the winning entry of the competition used filter bank CSP [7], which is one of the methods tried in this project.

2.3 Feature Extraction and Classification for Motor Imagery

Once the data is available, the next problem is turning the raw signal into features and training a model on them. This is harder than it sounds, because EEG signals vary a lot between people and even between sessions for the same person.

The standard way of looking at motor imagery is through event-related desynchronisation (ERD) and event-related synchronisation (ERS). ERD is a drop in power in a given frequency band during a task and ERS is a rise in power. For motor imagery, power in the alpha (8–15 Hz) and beta (15–30 Hz) bands usually drops over the motor cortex on the side opposite to the imagined hand [8]. The reference dataset paper computes this by filtering C3 and C4 between 8 and 30 Hz, squaring the signal to get power, averaging across trials and comparing against a baseline taken just before the task [4]. The same paper also benchmarks several classifiers on its own data for the left versus right MI problem, using the 8–30 Hz band and a 4 second window after the cue. Common Spatial Pattern with Linear Discriminant Analysis (CSP+LDA) [9] reached about 55.6%, Filter Bank CSP with a Support Vector Machine (FBCSP+SVM) [7, 10] about 57.6%, a Riemannian approach [11] about 61.2%, and the best method reported, a temporal window filter

bank combined with a Riemannian mean classifier, reached 72.21%. These numbers are a useful reminder that even with 30 channels and clean laboratory recordings, binary motor imagery classification is a difficult problem, and that accuracies in the 50–70% range are normal rather than poor.

Beyond CSP and SVM, deep networks are the other big family of methods. A survey by Xu and Xia walks through the network types that have been tried on EEG and the usual ways of preparing the signal for them [12]. Its main warning is about data size: deep networks learn well only from large amounts of data, and most EEG datasets are small. For a project that records its own data from a handful of people, a simple model like an SVM is therefore the sensible first choice.

EEG Conformer, proposed by Song and colleagues, mixes two ideas [13]. Convolution layers look for short local patterns in the signal, and an attention stage then relates those patterns to each other over the whole trial. The authors also show maps of which parts of the input the network relied on, and they report higher accuracy than CSP on the usual motor imagery benchmarks. It is a good candidate for later work, once there is enough data and computing power to train it.

Panachakel and Ramakrishnan used transfer learning for imagined speech [14]. Their network learns first from many people and is then tuned to a new person with only a little of that person's data, which shortens the setup for each new user. Accuracy still varied clearly from one person to the next, showing that differences between people remain a hard problem. The same problem appears in this project, where a model trained on one dataset is tested on data recorded from a different group of people with a different device.

Su and colleagues built a brain-computer interface for hand rehabilitation after stroke that uses two kinds of brain response at once: motor imagery, and the steady response the visual cortex gives to a flickering light [15]. Their model keeps updating itself as the patient's signals change over time. Using both signals gave higher accuracy than either alone, and patients' hand function improved, which shows that an EEG classifier can be of real clinical use.

2.4 Decoding Speech from EEG

A related line of research tries to work out what a person is saying, or only thinking of saying, from EEG alone. This is not part of the current objectives but it is the direction this project is expected to extend into, so the main findings are worth noting.

Krishna and colleagues recorded EEG while people spoke and found that a speech recogniser could use it, both when the audio was noisy and when there was no audio at all [16]. This mattered because it meant EEG on its own carries enough information about speech intent to be used for recognition. The experiments were done in a controlled lab, so how well this holds in noisy everyday conditions is still not known. In a follow-up study the same group tried to produce speech sound directly from EEG and managed a partial reconstruction of the spoken words [17]. The method depended on specific tasks and had to be trained separately for each person, which makes it hard to use on a new user without starting over.

EEG recorded during both imagined and spoken speech has also been used to classify the types of sounds being thought about or said [18]. This confirmed that imagined speech, where no sound is made at all, still leaves a pattern in EEG that can be detected, which is the basic assumption behind any silent communication system. On the listening side, the auditory EEG decoding challenge for ICASSP 2024 focused on working out what a person is listening to from their EEG [19]. The dataset and the evaluation method were released publicly, which helps the wider research community compare results fairly.

2.5 EEG for Cognitive and Mental State Assessment

Apart from decoding what a person intends to do or say, EEG is also widely used to measure how a person's mind is working at a given moment. This is the third objective of this project, which is to compute a cognitive score from EEG frequency bands.

Bird and colleagues used a low-cost headband and a set of statistical features to sort recordings into relaxed, neutral and concentrating states with machine learning [20], and found that these states can be told apart with good accuracy. The work looked at states in isolation, however, and did not follow how a changing cognitive state affects performance on an actual task.

Jawed and colleagues trained recurrent networks on unprocessed EEG to pick out people who learn best from visual material [21]. With 34 participants recorded at rest and while

learning, their best model, which combined an LSTM with convolution layers, was right about 94% of the time. What is useful here is that the models worked on raw EEG without heavy offline cleaning, which shows that real-time assessment from a continuous EEG stream is practical.

For driver drowsiness, Ming and colleagues used deep Q-learning, a reinforcement learning method that learns from feedback instead of from fixed class labels [22]. This is a useful idea because drowsiness builds up gradually and does not split neatly into classes. A more conventional system for the same problem used spectral entropy and power spectral density as features, which describe how the brain's electrical activity is spread across frequency bands, and fed them to boosting classifiers to detect signs of fatigue [23]. This second approach is closest to the method used in this project, where relative power in the delta, theta, alpha, beta and gamma bands is computed from the PSD and then combined into scores. In particular, the stress score used here is based on the ratio of relative high-beta power to relative alpha power, and the fatigue score on the ratio of combined theta and alpha power to beta power, both of which rest on the same band-power relationships these studies rely on. The limitation of such systems is that they are tuned for one specific state, so the thresholds and reference values have to be worked out again for any new measure.

2.6 Summary of the Literature Survey

The studies reviewed above map onto the three objectives of this project as follows.

Building an EEG database for motor imagery. Wearable dry-electrode systems, including ear-mounted ones, can record usable EEG with a small number of channels and very little setup [2,3], and the open stroke-patient motor imagery dataset provides a well-documented left and right hand MI recording protocol that can be followed when collecting new data [4]. The BCI Competition IV 2b dataset adds healthy subjects recorded with only three channels at 250 Hz, which is much closer to a small wearable headset [5].

Designing a motor imagery classifier. ERD and ERS analysis of C3 and C4 in the 8–30 Hz band is the accepted way to look at motor imagery, and CSP, filter bank CSP, LDA, SVM and Riemannian methods form the standard set of classifiers, with reported accuracies for left versus right MI ranging from about 55% to 72% [4]. Deep learning offers further gains [12, 13], transfer learning helps when moving between users [14], and

adaptive models help when a user's signals drift over time [15].

Computing a cognitive score from frequency bands. Relative power in the EEG frequency bands, together with PSD and entropy features, can be used to separate mental states and to track attention and fatigue [20–23].

Three gaps stand out from this review. First, almost all of the motor imagery results reported above come from research-grade systems with 20 or more channels, while a practical wearable device may have only a few; how much accuracy is lost in that reduction is not well documented. Second, models are usually trained and tested on the same dataset and the same recording device, so very little is known about what happens when a model trained on one dataset is applied to data recorded elsewhere with different hardware. Third, work on motor imagery decoding and work on cognitive state monitoring have largely been carried out separately, even though both are computed from the same recording. This project addresses these three points by recording left and right motor imagery on a four-channel dry headset, training classifiers on the public dataset and testing them on the recorded data, and computing a cognitive score from the same signal within a single system.

Chapter 3

System Overview

This project builds a system that uses EEG signals recorded from the scalp to do two things. The first is to tell whether a person is imagining a movement of the left hand or the right hand. The second is to find out the mental condition of the person at that moment. Both outputs come from the same recording. The system takes the raw brain signals, cleans them, pulls out useful information from them, and then uses a machine learning model for the left versus right decision and a set of frequency band calculations for the cognitive score. No audio is recorded at any stage.

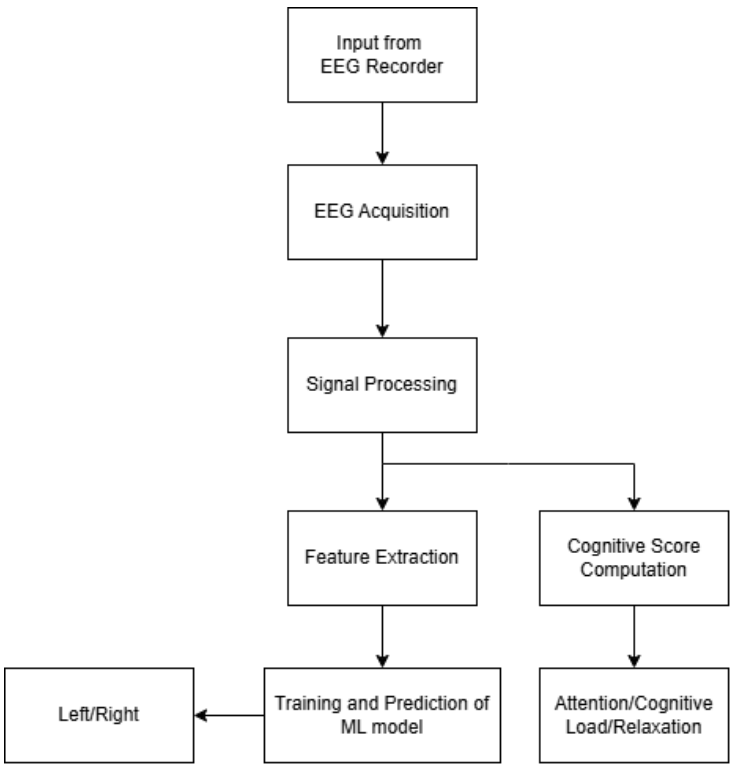


Figure 3.1: EEG Based Cognitive Assessment Overview

Figure 3.1 illustrates the overall architecture of the proposed system. The first step is collecting the data. The person wears an EEG headset with electrodes placed on the scalp at standard positions. During the recording session the person is shown an instruction

and is asked to imagine gripping an object with the left hand or the right hand, while the EEG signals are being recorded. The task is only imagined, so no actual movement is made and no sound is produced. To know which task was being performed at each point in the recording, markers are placed in the recording software at the start of every trial. These markers are what label the data, and they replace the need for any audio reference.

The filtered EEG is then sent to the band analysis stage. Here the signal is divided into the standard frequency bands, which are delta, theta, alpha, beta and gamma, to study the brain activity. The bands matter for both halves of the system. For motor imagery, the alpha and beta bands over the motor cortex are the ones that change when a movement is imagined. For the cognitive score, the balance of power between the bands is what indicates whether the person is attentive, under load, relaxed, stressed or tired.

The next step is feature extraction, which means taking out useful information from the EEG signals. For the motor imagery path, the two motor cortex channels C3 and C4 are filtered to the 8–30 Hz range, their power is computed, and the change in power with respect to a baseline taken just before the task is calculated. Statistical measures such as the mean, median, skewness and kurtosis of each channel are then worked out, along with the ratio of power between C3 and C4 and the band power over 8–30 Hz. These values together form a set of numbers that represent what the motor cortex was doing during that trial. For the cognitive path, the power spectral density is computed and the relative power in each band is obtained by dividing the power in that band by the total power.

The machine learning model is trained using the features extracted from the EEG data along with the labels that say whether the trial was a left-hand or a right-hand imagination. The model looks at the patterns in the features and tries to learn which pattern belongs to which side. Once training is done, the model can take a new EEG trial and predict whether the person imagined the left or the right hand. In parallel, the relative band powers are combined into ratios to produce the cognitive outputs, which are attention, cognitive load, relaxation, stress and fatigue.

The output of the system is therefore the predicted side, left or right, along with the cognitive score. The predicted side shows which hand the person was imagining, and the cognitive score shows how attentive, stressed or tired the person is at that moment. Both outputs come from the same EEG recording setup, which keeps the system simple and

practical to use.

Overall this system takes EEG signals as input, processes them, and uses machine learning and frequency band analysis to give two useful outputs. It works with a simple wearable headset and needs nothing besides the EEG signal itself, which makes it a practical approach for real world use.

Chapter 4

System Hardware

This chapter goes through the hardware used in the project and why each part was picked. There are four main parts: the GELO headset that picks up the EEG, the Raspberry Pi 5 that is meant to run the processing, a metal case with fans that keeps the Pi cool, and a small touch screen that shows the interface. The EEG recorder is covered first, since the choice of recorder decides most of what the rest of the system can do.

4.1 EEG Recorder

An EEG recorder is a device that measures the tiny voltages on the scalp through a set of electrodes and sends them to a computer. Recorders differ in how many electrodes they have, how the electrodes touch the skin, how comfortable they are, and whether they need a cable. The two main families are described below.

4.1.1 Wet Electrode EEG System

In a wet system a gel or saline solution is put between each electrode and the skin. The liquid lowers the contact resistance, so the recorded signal is clean and steady. This is why hospitals and research labs use wet caps, usually with the electrodes laid out according to the international 10–20 placement scheme [24], which names each position by its distance from fixed points on the skull.

For this project a wet system was not a good fit. Putting gel on every electrode takes time before each session, the gel dries out over a long recording, the hair has to be washed afterwards, and the cap itself needs cleaning. All of this goes against the aim of a headset that a person can put on and use in daily life without help.

4.1.2 Dry Electrode EEG System

A dry system presses the electrodes directly against the skin with no gel at all. Setup takes a minute or two, the person can wear the device without help, and nothing needs cleaning afterwards. The trade-off is a higher contact resistance, so the signal is somewhat noisier and more sensitive to movement. Dry electrodes appear in several shapes of device,

such as headbands and headphones. A headphone design was chosen here, because it sits firmly over the ears and the top of the head, where the motor cortex electrodes need to be, and it is comfortable for longer sessions. This led to the GELO BCI headset.

4.1.3 GELO BCI Headset

The GELO BCI headset, shown in Figure 4.1, looks and works like an ordinary pair of wireless headphones, but it has dry electrodes built into the headband and ear cups. It records four EEG channels, named A1, A2, C3 and C4, against a reference electrode at Cz, with two further electrodes acting as ground. The names follow the 10–20 scheme [25]. These positions are the main reason the device suits this project. C3 lies over the left motor cortex and C4 over the right, and these are the two places where the activity changes when a person imagines moving a hand, so they carry the information needed for the left and right decision. A1 and A2 sit on the ears, where there is very little brain activity. Each motor channel is read against its ear electrode, as A1–C3 and A2–C4, so that noise reaching both electrodes of a pair is largely cancelled. All four channels are also used for the band powers behind the cognitive score.

The headset samples each channel 250 times a second, which is more than enough for the 0.5–35 Hz range studied here, and sends the data over Bluetooth Low Energy, so the person is not tied to the computer by a cable. Since it is also a pair of headphones, it has speakers and a microphone. Neither is used in this work, as no sound is recorded, but the speakers could later play an audio cue at the start of each trial. The main specifications are listed in Table 4.1.

Table 4.1: Main specifications of the GELO BCI headset [25]

Item	**Value**
EEG channels	4 (A1, A2, C3, C4)
Reference and ground	Cz reference, two ground electrodes
Electrode type	Dry, gold-plated spring pins
Sampling rate	250 Hz per channel
Read mode	Bipolar (A1–C3, A2–C4)
Wireless link	Bluetooth Low Energy 5.0
Audio	Two 40 mm speakers, 20 Hz – 20 kHz
Bluetooth audio profiles	HFP, HSP, A2DP



Figure 4.1: GELO BCI Headset

4.2 Raspberry Pi 5

The Raspberry Pi 5, shown in Figure 4.2, is the small computer chosen to run the whole system on its own once development is finished. Its job is to receive the EEG from the headset over Bluetooth, filter it, compute the features, run the classifier and the cognitive score, and draw the interface on the screen.

At its heart is a Broadcom BCM2712 chip with four Arm Cortex-A76 cores running at up to 2.4 GHz [26]. The processing in this project is light by modern standards: a few filters, a Welch spectrum and a linear SVM on a handful of features every fraction of a second, so this processor has plenty of headroom. The board has Bluetooth built in, which is how the headset connects, along with Wi-Fi, Ethernet, USB 3.0 ports and two micro-HDMI outputs, one of which drives the display. The 40-pin header can be used later to send the left or right decision to a motor driver or a robot. Its main specifications are given in Table 4.2.

Table 4.2: Main specifications of the Raspberry Pi 5 relevant to this project [26]

Item	**Value**
Processor	Broadcom BCM2712, 4 × Cortex-A76, 2.4 GHz
Memory	4 GB or 8 GB LPDDR4X
Wireless	Wi-Fi (2.4 and 5 GHz), Bluetooth 5.0 / BLE
Wired	Gigabit Ethernet, 2 × USB 3.0, 2 × USB 2.0
Display output	2 × micro-HDMI
Expansion	40-pin GPIO header, PCIe 2.0 ×1
Other	Real-time clock, power button

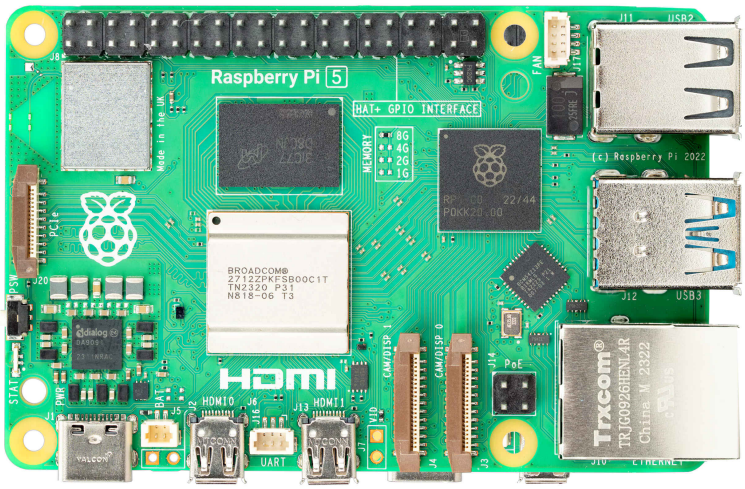


Figure 4.2: Raspberry Pi 5

4.3 Aluminum Heat Sink Case with Dual Fans for Raspberry Pi 5

The Raspberry Pi 5 gets hot when it works without a break, and once its temperature passes a set limit it slows its own clock to protect itself. For a system that has to process EEG continuously for a whole session, that slowdown would make the output lag. The case shown in Figure 4.3 prevents this.

The case is machined from aluminium, a metal that conducts heat very well, so the whole body of the case acts as one large heat sink that draws heat away from the chip. Two small fans on top blow air across it to remove that heat faster. The metal body also shields the board from knocks and dust, which matters because the unit is meant to be carried around with the headset rather than sit on a lab bench.

4.3.1 Features of Aluminum Heat Sink

- The aluminium body itself works as a large passive heat sink
- Two fans add airflow when the processor is under steady load
- Keeps the Pi below the temperature at which it slows down
- Gives the board a solid, closed housing for portable use

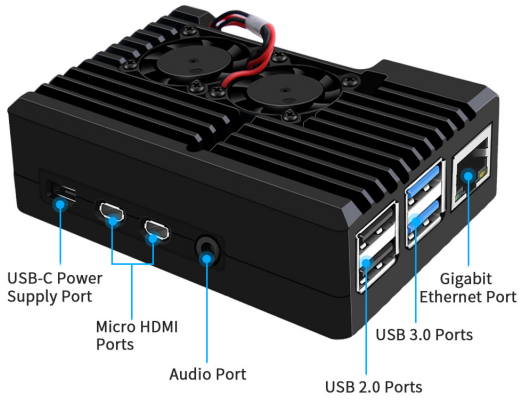


Figure 4.3: Aluminum Heat Sink Case with Dual Fans for Raspberry Pi 5

4.4 Waveshare 7-Inch LCD Display

The 7-inch Waveshare touch screen [27], shown in Figure 4.4, is how the person using the system sees what is going on. During a recording it shows the left and right arrows the subject has to follow, the live EEG traces, and the status of the electrodes. After processing it shows the predicted side and the cognitive scores.

The screen connects to the Raspberry Pi through HDMI for the picture and USB for the touch input, and it is supported by Raspberry Pi OS. Because it is a touch screen, the session can be started, calibrated and stopped with a finger, so no keyboard or mouse is needed. Its small size keeps the whole set-up portable, which is useful for demonstrations and for recording outside the lab.

4.4.1 Features of Waveshare LCD Display

- 7-inch screen, large enough to read the scores from a normal distance
- HDMI picture input and USB touch input, both supported by the Pi
- Touch control, so no keyboard or mouse is needed during a session
- Light and compact, so the system can be carried with the headset

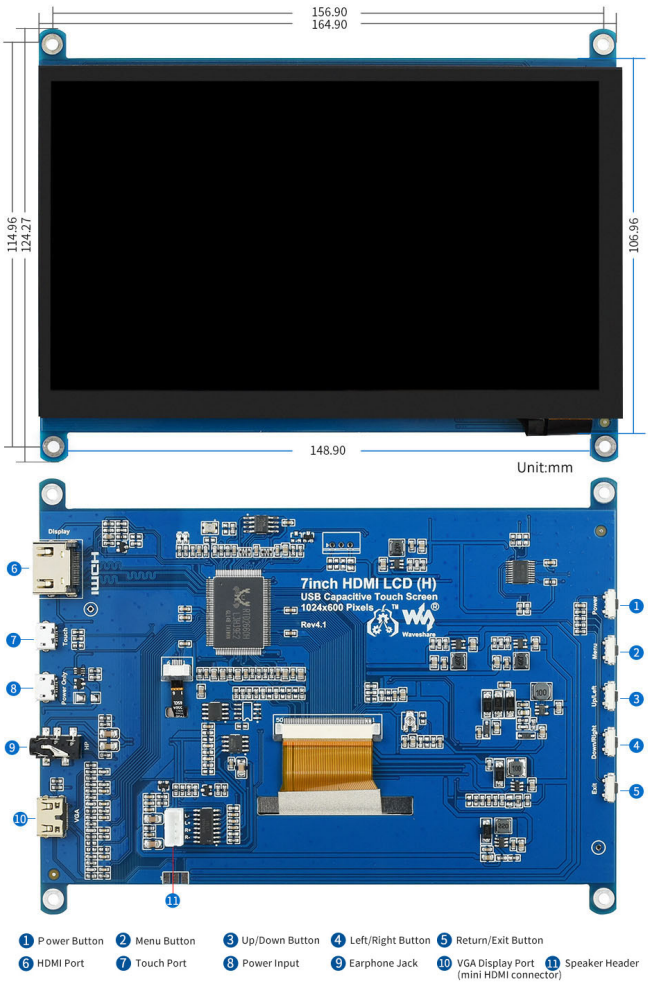


Figure 4.4: Waveshare 7-Inch LCD Display

Chapter 5

System Software

This chapter explains the software setup used to develop the proposed EEG-based motor imagery classification and cognitive assessment system. It includes the software environment, the GUI software, the EEG signal collection process, and the libraries used for signal processing, visualization, machine learning and cognitive score calculation. The system combines real-time EEG data collection using the Capsule GUI with Python-based preprocessing and classification methods to perform EEG signal analysis in a reliable way.

5.1 Software Requirements

The proposed system uses the Capsule GUI software for real-time EEG signal acquisition, electrode impedance checking, calibration, signal display and marker placement. All recording and processing reported in this work is carried out on a Windows 11 computer. The Raspberry Pi 5 described in the previous chapter is the intended deployment platform, so that the complete system can later run as a standalone unit, but the results presented in this report were obtained on the Windows setup.

The recording and analysis set-up is summarised in Table 5.1, and the Python libraries, with the job each one does in this project, are listed in Table 5.2.

Two plotting libraries are used on purpose. Matplotlib gives neat static figures for analysis and for this report, but it is slow to redraw, so the live trace that has to update many times a second is drawn with PyQtGraph instead.

5.2 Flowchart

The system shown in Figure 5.1 starts with the person putting on the EEG headset and the headset being paired with the computer over Bluetooth. The resistance of each of the four electrodes is then checked, since a poor contact makes the rest of the recording useless. Two short calibration steps follow. In the first, the subject keeps the eyes closed for 30 seconds so that the individual alpha peak frequency can be found, as alpha activity is strongest with the eyes closed. In the second, the subject looks at a fixed point for 30

Table 5.1: Software set-up used in this project

Item	**Used for**
Windows 11	All recording and analysis reported here
Raspberry Pi OS	Planned stand-alone version of the system
Capsule GUI	Connecting the headset, checking electrode resistance, calibration, live view, placing trial markers
Python 3	Conversion of recordings, feature calculation, SVM training, cognitive score and the application window
MATLAB	ERD/ERS and PSD of the stroke and BCI Competition IV 2b datasets, the ERD% SVM on the 2b data (`fitcsvm`) and the report figures

Table 5.2: Python libraries and what each one does in this project

Library	**Job in this project**
NumPy	Holds the EEG samples as arrays; sums spectrum bins into bands; computes the ratios
SciPy	Butterworth band-pass and notch filters, Welch spectra
Pandas	Reads the CSV files and keeps track of which trial belongs to which hand
h5py	Opens the `.h5` files saved by the Capsule software
MNE-Python [28]	Channel handling and cutting the recording into trials around each marker
Scikit-learn [29]	Feature scaling, SVM training and accuracy checks
Matplotlib	Static plots: PSD, ERD curves, band bars and the figures in this report
PyQtGraph	Fast-refreshing live EEG trace during a session
CustomTkinter	The application window: sidebar, cards, buttons and dialogs
PyTorch (optional)	Kept for possible deep learning models later

seconds so that the median power in each frequency band can be stored as a baseline. Later readings are compared against this baseline rather than against fixed values, which makes the scores specific to that person.

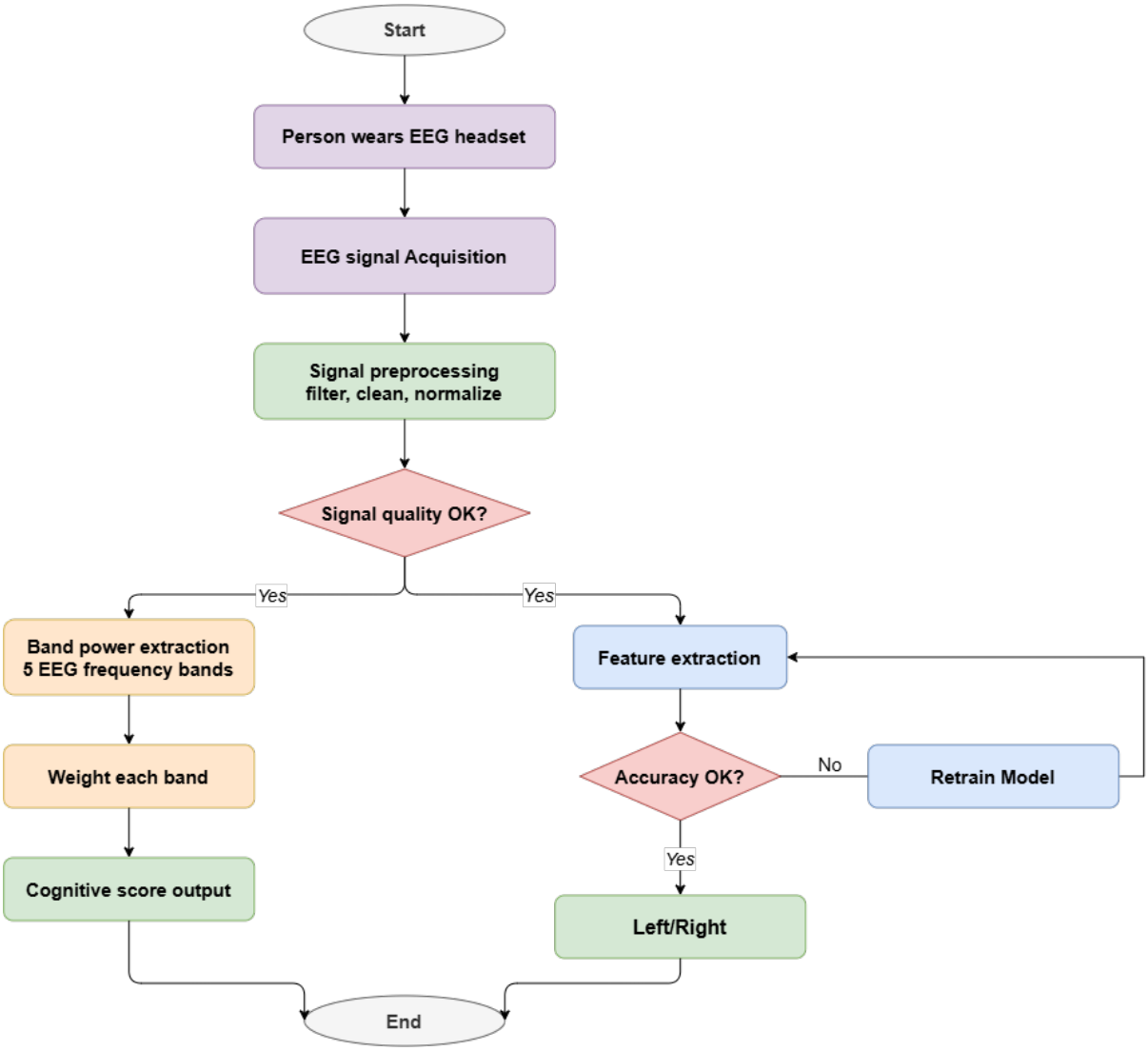


Figure 5.1: Flowchart of EEG based Cognitive Assessment Model

Once calibration is complete, the recording begins and the subject performs the motor imagery trials. A marker is placed at the start of each trial so that every segment of the recording is tied to a known label, either left-hand or right-hand imagination. The session is then saved in `.h5` format and converted to CSV using a Python script for further analysis. No audio is recorded at any point.

If the signal is too noisy or unclear, the recording is done again from the beginning. Only when the signal quality is good enough does the system move forward. After passing the quality check, the pipeline splits into two separate paths that run independently.

The first path handles cognitive assessment. Here the EEG signal is broken down into the five frequency bands, which are delta, theta, alpha, beta and gamma. The power of each band is calculated and divided by the total power to give its relative power, which removes the effect of the overall signal level changing as electrode contact improves or worsens. These relative powers are then combined into ratios to give the cognitive outputs. Attention, cognitive load and relaxation are read from the device SDK, while stress is obtained from the ratio of relative high-beta power to relative alpha power, and fatigue from the ratio of combined theta and alpha power to beta power. Each ratio is compared against the person's own reference value and mapped onto a score between 0 and 100, so the result tells how mentally active or tired the person is at that moment.

The second path handles motor imagery classification. In this path the C3 and C4 channels are taken out and filtered to the 8–30 Hz band, a fixed window of 4 seconds is cut from the start of each imagination, and the power and the change in power relative to the baseline are computed. Features such as the mean, median, skewness and kurtosis of each channel, the ratio of power between C3 and C4, and the band power over 8–30 Hz are then extracted. These features are used to train a Support Vector Machine. After training, the model accuracy is checked. If the accuracy is not satisfactory, the process goes back to the feature extraction stage, the feature set or the kernel is adjusted, and the model is retrained. This loop continues until the accuracy reaches an acceptable level. Once the accuracy is good, the model is used to predict left or right from new EEG recordings.

Both paths finish at the end node, giving two outputs from a single EEG recording setup, which are the cognitive score and the predicted side.

Chapter 6

Results

This chapter presents the experimental results obtained from the proposed EEG-based system. The work has two parts and the results follow the same split. The first part deals with motor imagery, where the aim is to find out whether EEG recorded from a four-channel dry headset carries enough information to tell a left-hand imagination apart from a right-hand one. The second part deals with cognitive assessment, where the aim is to compute attention, cognitive load, relaxation, stress and fatigue from the frequency bands of the same signal. Results are reported in the order the work was carried out, starting with the reproduction of the stroke dataset analysis, then the same analysis and a set of classifiers on the BCI Competition IV 2b dataset of healthy subjects, then the analysis of the data recorded in this project and its classifier, and finally the cognitive scoring system.

6.1 Evaluation Aspects

For the motor imagery part, the EEG signals recorded during imagined left-hand and right-hand grasps are compared in the time domain, through event-related desynchronisation and synchronisation (ERD/ERS), and through power spectral density (PSD). These analyses check whether the two classes show any consistent difference that a classifier can use. For the cognitive part, the relative power in each frequency band is tracked over a session and converted into scores, and the stability of the recording itself is checked through the electrode impedance.

6.1.1 Cases of Comparison

The following comparison cases were studied during the experimental analysis:

1. ERD/ERS of the stroke reference dataset, reproduced to validate the analysis method

2. ERD/ERS, PSD, C3/C4 power ratio and SVM accuracy on the BCI Competition IV 2b dataset of healthy subjects

3. ERD/ERS of the data recorded in this project, compared against the reference
4. PSD of C3 and C4 for imagined left versus imagined right
5. Classification accuracy of different SVM feature sets and kernels
6. Band power and cognitive scores recorded over a complete session

6.1.2 Parameters Under Study

The main parameters considered in this analysis are:

1. EEG channels: C3, C4, A1 and A2, with C3 and C4 as the motor cortex pair
2. Motor imagery classes: imagined left-hand grasp and imagined right-hand grasp
3. Frequency range: 8–30 Hz for motor imagery, 1–30 Hz for cognitive scoring
4. Features: band power, C3/C4 power ratio, mean, median, skewness and kurtosis
5. Cognitive outputs: attention, cognitive load, relaxation, stress and fatigue

6.2 Experimental Setup

The experiment was carried out using the GELO BCI headset with four EEG channels and one reference electrode, sampling at 250 Hz. The acquisition was done through the Capsule GUI, shown in Figure 6.1. Before every session the headset was paired over Bluetooth, the electrode resistances were checked, and the two calibration steps described in Chapter 5 were performed.

During each recording session the subject was seated comfortably and asked to imagine gripping an object with the left hand or the right hand when prompted. The movement was only imagined, so no actual movement was made. Each imagination was marked in real time using a marker, which was recorded alongside the EEG so that the start and end of every trial could be identified later. Each recorded file contains five marked repetitions of one class. The session was saved in `.h5` format and converted to CSV for processing.

For post-recording analysis the CSV files were loaded and sorted by class based on their filenames. The marker column was used to extract the active imagination windows as individual trials. Since the shortest trial across the recordings was close to four seconds,

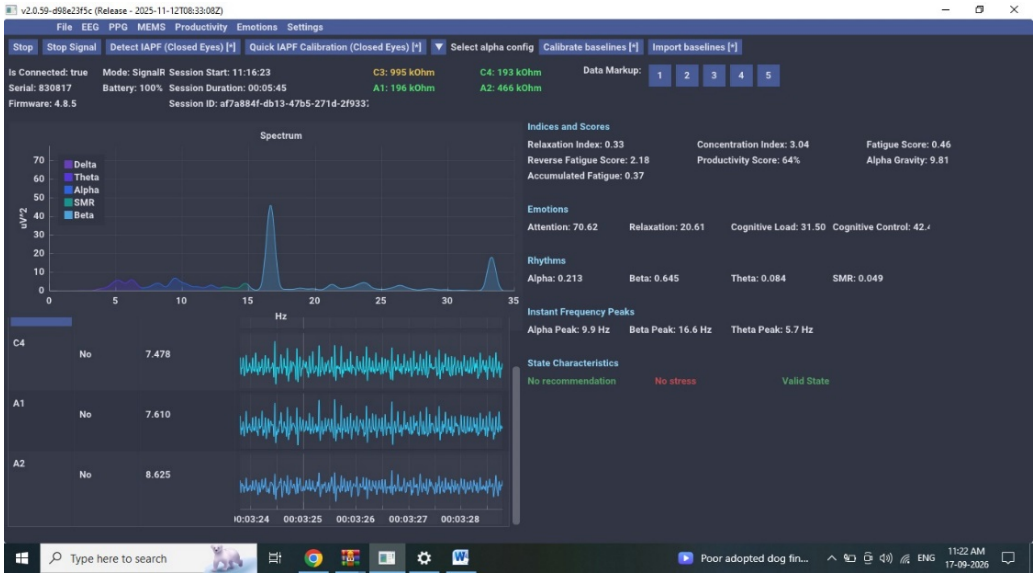


Figure 6.1: EEG signal acquisition using the Capsule GUI

a fixed window of four seconds from the onset of each imagination was used so that every trial contributes the same number of samples. The signal supplied by the device is already filtered, so no further filtering was applied before the feature calculation except the 8–30 Hz band limiting used for the ERD and PSD analysis. PSD was estimated using Welch's method [30] with a window size of 256 samples at $F_s = 250$ Hz. Out of all the recordings made, 50 right-hand trials and 22 left-hand trials were found usable, the rest being discarded because of poor electrode contact or movement during the trial.

6.3 Reproduction of the Stroke Dataset Analysis

Before working on the recorded data, the ERD/ERS analysis of the reference dataset [4] was reproduced. This was done to confirm that the analysis method used in this project gives the same result as the published work before it is applied to new data. Event-related desynchronisation refers to a drop in EEG power in a given frequency band during a task, and event-related synchronisation refers to a rise [8]. For motor imagery, power in the 8–30 Hz range is expected to fall over the hemisphere opposite to the imagined hand.

The C3 and C4 signals were extracted, filtered between 8 and 30 Hz using a second-order zero-phase Butterworth filter, squared to obtain power, averaged across trials, and smoothed. The change was computed as

$$\text{ERD/ERS}\ (\%) = \frac{X - M}{M} \times 100 \tag{6.1}$$

where X is the power at a given instant and M is the average power during the one second preparation period before the task. Figure 6.2 shows the result obtained for subject 8 of the dataset. During left-hand imagery the C4 trace separates clearly from C3 and falls through the middle of the trial, which is the expected contralateral response. During right-hand imagery the two traces stay much closer together. This matches the behaviour reported in the dataset paper and confirms that the processing chain is correct.

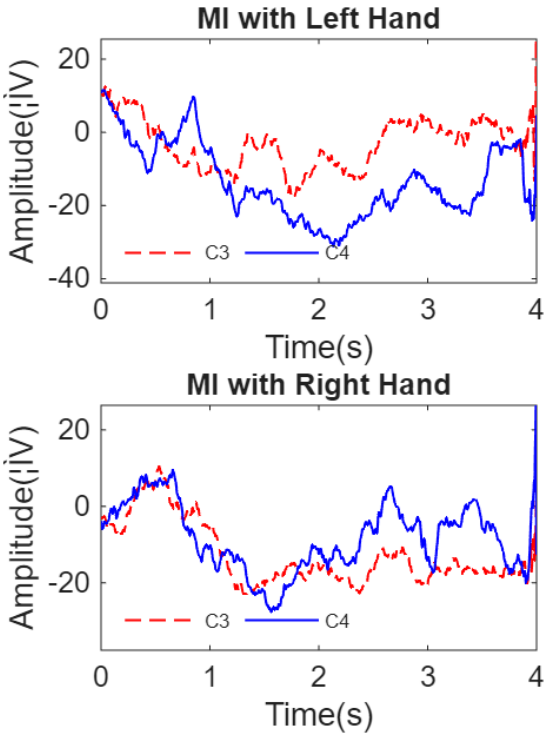


Figure 6.2: ERD/ERS of C3 and C4 for subject 8 of the reference dataset [4], reproduced using the analysis method of this project

6.4 Results on the BCI Competition IV 2b Dataset

The stroke dataset turned out to be a hard place to start. Even the published methods, using all 30 channels, reach only 55 to 72 per cent on it [4], so a low score there cannot tell us whether our own processing is right or wrong. A second reference was needed, one that is closer to the GELO headset. The BCI Competition IV 2b dataset, called the BCI 2008 dataset in this report and described in Section 2.2, fits this need well: the subjects are healthy, the sampling rate is 250 Hz, and the electrodes sit at C3 and C4 just like on the headset. Every step in this section uses the same method that was used for the stroke patients, so that the two can be compared directly.

6.4.1 How the Dataset Was Used

The dataset was downloaded as 18 MATLAB files, one training file and one evaluation file for each of the nine subjects. The training file holds sessions 1 to 3 and the evaluation file holds sessions 4 and 5. Each file contains the continuous EEG of the three channels (C3, Cz and C4) and the three EOG channels, the sample number at which every trial starts, the class of every trial (1 for left hand, 2 for right hand), and a flag that the dataset authors set on trials they found to contain artefacts. The labels of the evaluation sessions were kept secret during the competition and released afterwards, so all five sessions could be used here. In total there are 6520 trials, 3260 for each hand.

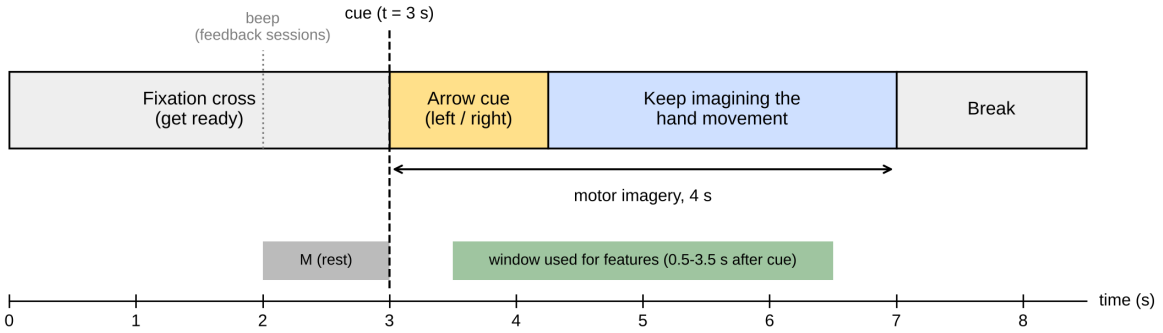


Figure 6.3: Timing of one trial in the BCI Competition IV 2b dataset. M is the rest period used as the reference for ERD/ERS, and the green bar is the window used for the classifier features

Figure 6.3 shows what happens in one trial. A fixation cross appears at the start, an arrow pointing left or right appears at 3 s, and the subject keeps imagining the movement of that hand until about 7 s, after which there is a short break. One detail in the files is easy to miss and cost a full round of work in this project. The position stored for each trial is the start of the trial and not the moment of the cue, which comes 3 s later. The first classifiers trained here cut their windows from the trial start, so they looked only at the fixation period and all of them came out at exactly chance level. Once the windows were moved 3 s forward to the cue, the accuracies jumped to the values reported below. The same kind of mistake, an event marker read in the wrong way, is something to check first whenever a classifier gives exactly 50 per cent.

Only C3 and C4 were used for the ERD/ERS and PSD analysis, since those are the two

motor cortex channels the GELO headset has. The classifiers were run with all three channels and again with only C3 and C4, and leaving out Cz lowered the cross-subject accuracy by less than half a percentage point. Trials flagged as artefacts were kept in the main analysis. The classifiers were also re-run with these trials removed, and the result changed by less than half a percentage point, so the choice does not affect any conclusion.

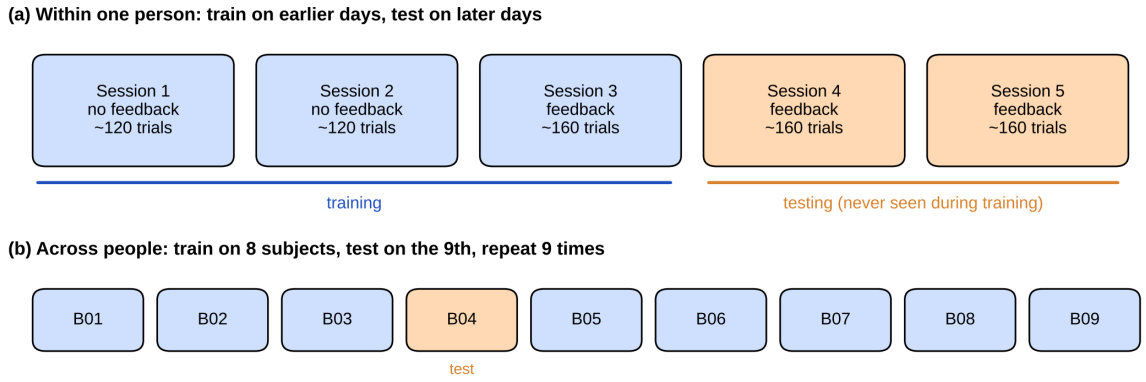


Figure 6.4: The two ways the classifiers were tested on the BCI Competition IV 2b data

Two ways of testing were used, as shown in Figure 6.4. In the first, a separate model is trained for each person on sessions 1 to 3 and tested on sessions 4 and 5, which were recorded on later days. This is the official split of the competition. In the second, the model is trained on eight people and tested on the ninth, and this is repeated so that every person is tested once. The second test is the one that matters most for this project, because it shows whether a model can work on a new person with no training data from that person, which is what the GELO headset would need. In both cases the test trials are never used for training, for scaling the features or for choosing any setting. The complete processing chain is shown in Figure 6.5.

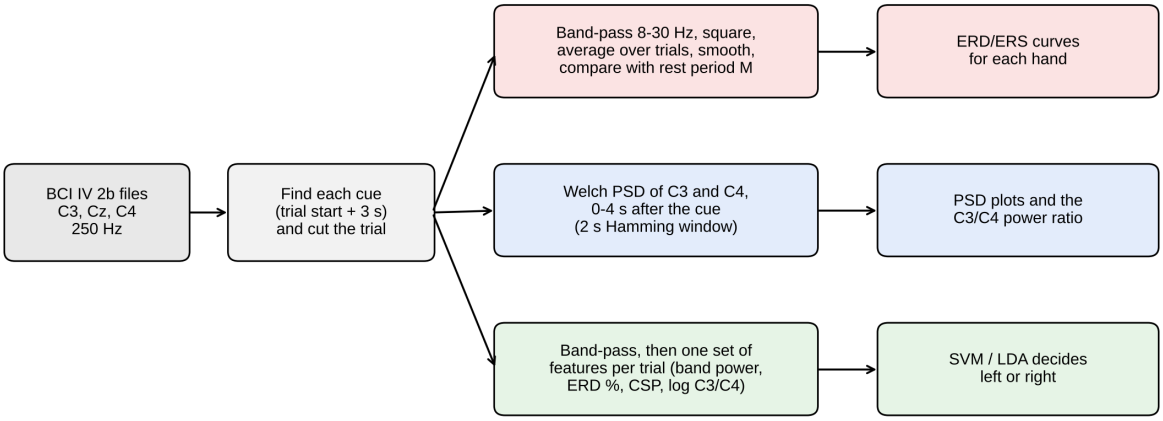


Figure 6.5: Processing chain used for the BCI Competition IV 2b data

6.4.2 ERD/ERS of the BCI Competition IV 2b Subjects

The ERD/ERS was computed in exactly the same way as for the stroke patients. C3 and C4 were filtered between 8 and 30 Hz with a second-order zero-phase Butterworth filter, squared to give power, averaged over all trials of the same hand, smoothed with a 100-sample moving average (0.4 s), and converted to a percentage change using the same equation as before, with M taken as the average power in the one second just before the cue. Two small changes were made so that the whole trial can be seen. The plot covers the full 8 s of the trial with the cue marked, and the filter was run on the trial with an extra half second on each side, which was then cut off, so that the start and end of the curve are not distorted by the filter.

Figure 6.6 shows the result for subject B04, the subject with the clearest pattern in the dataset. Before the cue both channels stay close to zero, as they should, since nothing has happened yet. During left-hand imagery, C4 falls to about -50 per cent within one second of the cue and stays well below C3 for the rest of the imagery period, while C3 dips briefly and then returns close to the baseline. During right-hand imagery the picture is reversed: C3 falls to about -60 per cent, and C4 not only stays up but rises to about $+45$ per cent. This is the textbook pattern described in Chapter 2 [8]: imagining one hand lowers the 8–30 Hz power over the opposite side of the head, and the side that is not involved can even become more synchronised than at rest.

Table 6.1 gives the average change for all nine subjects. The value in bold is the channel

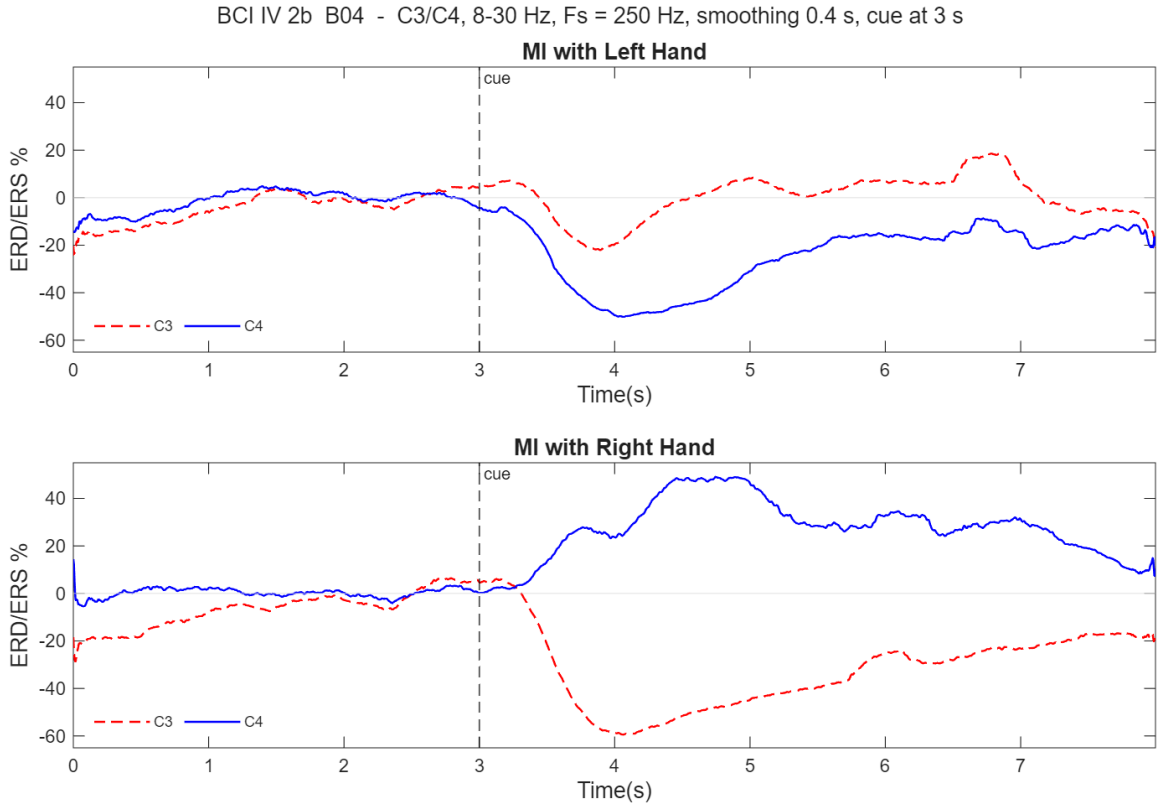


Figure 6.6: ERD/ERS of C3 and C4 for subject B04 of the BCI Competition IV 2b dataset, 8–30 Hz, $F_s = 250$ Hz, 370 trials for each hand. The dashed line marks the cue at 3 s

Table 6.1: Mean ERD/ERS (%) from 0.5 to 3.5 s after the cue for every BCI Competition IV 2b subject

Subject	**Left-hand imagery**		**Right-hand imagery**	
	C3	**C4**	**C3**	**C4**
B01	−15.4	**−40.0**	**−24.3**	−15.7
B02	−13.7	**−31.8**	−17.1	−23.4
B03	−14.5	**−25.3**	**−20.4**	−17.3
B04	−1.2	**−31.1**	**−42.2**	+33.8
B05	−27.6	**−30.3**	**−35.1**	−16.2
B06	−10.1	**−30.9**	**−29.9**	−16.6
B07	−2.4	**−17.0**	**−27.8**	+6.7
B08	−7.7	**−23.8**	**−37.6**	+52.2
B09	−18.7	−16.2	**−45.6**	−2.6
Median	−13.7	−30.3	−29.9	−15.7

opposite to the imagined hand whenever that channel dropped more. This happens in 16 of the 18 cases. The two exceptions are B02 during right-hand imagery and B09 during left-hand imagery, and in both of them the two channels dropped by similar amounts rather than the wrong channel dropping clearly. The plots for all nine subjects are collected in Figure 6.7.

There is one clear difference from the data recorded with the GELO headset. In BCI Competition IV 2b both channels fall below the baseline after the cue, which is a true desynchronisation, while in the recorded data both channels mostly rose. In 2b the second before the cue is a calm fixation period, so M is a real rest level. This supports the explanation given later in Section 6.5 that the baseline in the recorded sessions is not a true rest period.

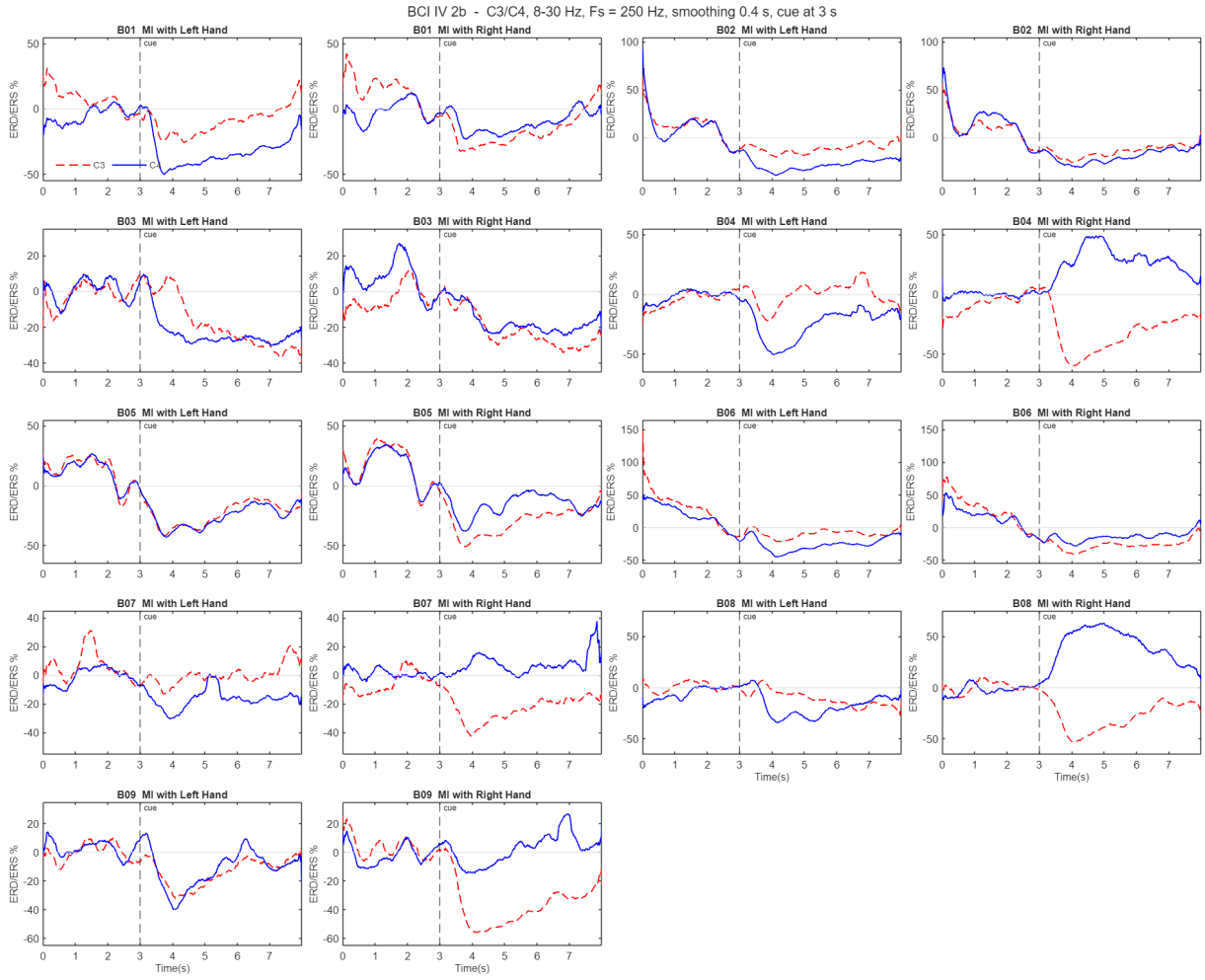


Figure 6.7: ERD/ERS of C3 and C4 for all nine subjects of the BCI Competition IV 2b dataset

6.4.3 Power Spectral Density of Subject B04

The PSD was computed with Welch's method [30] on C3 and C4 for the 4 s after the cue, using 2 s Hamming windows with 50 per cent overlap, which gives a resolution of 0.5 Hz. The spectra of all trials of the same hand were averaged. The C3/C4 ratio printed above each figure is the power between 8 and 30 Hz at C3 divided by the power in the same band at C4. Figures 6.8 and 6.9 use the same layout as the PSD figures of the recorded data in Section 6.5, with the two channels on top and the overlay below.

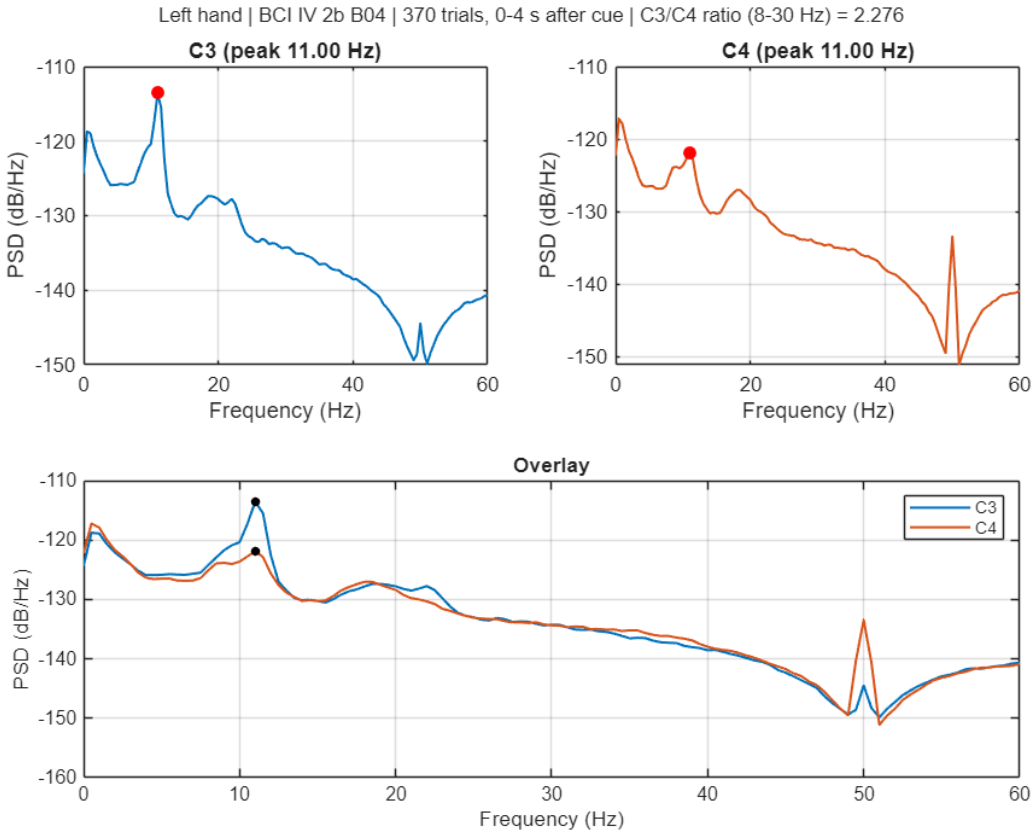


Figure 6.8: Power spectral density of C3 and C4 for imagined left-hand movement, subject B04 of the BCI Competition IV 2b dataset

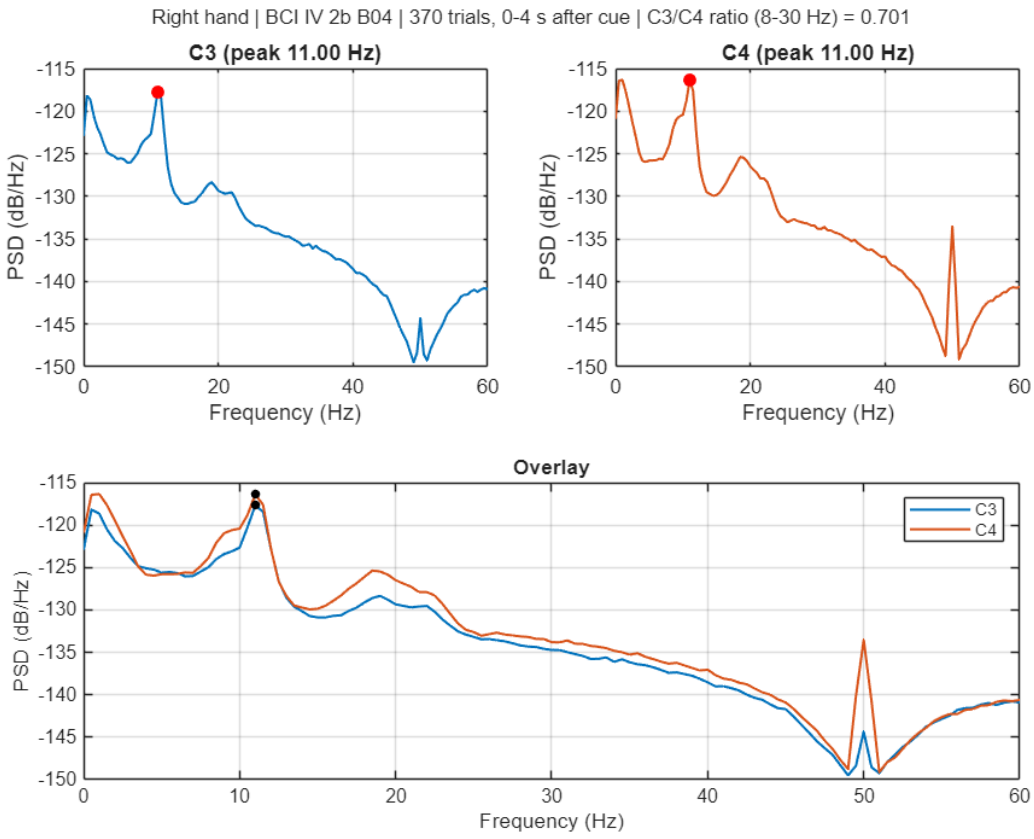


Figure 6.9: Power spectral density of C3 and C4 for imagined right-hand movement, subject B04 of the BCI Competition IV 2b dataset

Both channels have a clear peak at 11 Hz for both hands, which is the mu rhythm of the motor cortex, and a smaller bump around 18 to 22 Hz in the beta band. The peak frequency is the same in all four plots, so, just as with the recorded data, the peak frequency does not tell the two hands apart. What changes is which channel is stronger. During left-hand imagery the C3 peak stands about 8 dB above the C4 peak, because the power at C4, over the right side of the head, has dropped. During right-hand imagery the order flips and C4 is slightly above C3 across the mu and beta bands. As a result the C3/C4 ratio is 2.28 for the left hand and 0.70 for the right hand. The sharp feature at 50 Hz comes from mains interference and the notch filter used during recording, and it lies well outside the 8–30 Hz band used in this work.

6.4.4 The C3/C4 Power Ratio as a Left and Right Detector

The same ratio was computed for every subject, and the result is given in Table 6.2 and Figure 6.10. The reasoning behind it is simple. Imagining the left hand lowers the power at C4, so C3/C4 goes up. Imagining the right hand lowers the power at C3, so C3/C4 goes down. Using C4/C3 instead carries exactly the same information, with the direction reversed.

Table 6.2: C3/C4 power ratio (8–30 Hz, 0–4 s after the cue) and single-trial accuracy of the ratio for each BCI Competition IV 2b subject

Subject	**Left hand**	**Right hand**	**Left higher?**	**Accuracy, sessions 4–5**
B01	0.96	0.63	Yes	60.9%
B02	0.91	0.83	Yes	55.7%
B03	1.21	1.08	Yes	56.3%
B04	**2.28**	**0.70**	Yes	**96.6%**
B05	0.98	0.71	Yes	69.1%
B06	1.30	0.87	Yes	79.1%
B07	1.14	0.80	Yes	69.4%
B08	0.92	0.31	Yes	90.6%
B09	1.05	0.58	Yes	83.8%
Mean			9 of 9	73.5%

For every one of the nine subjects the ratio is higher for the left hand than for the right

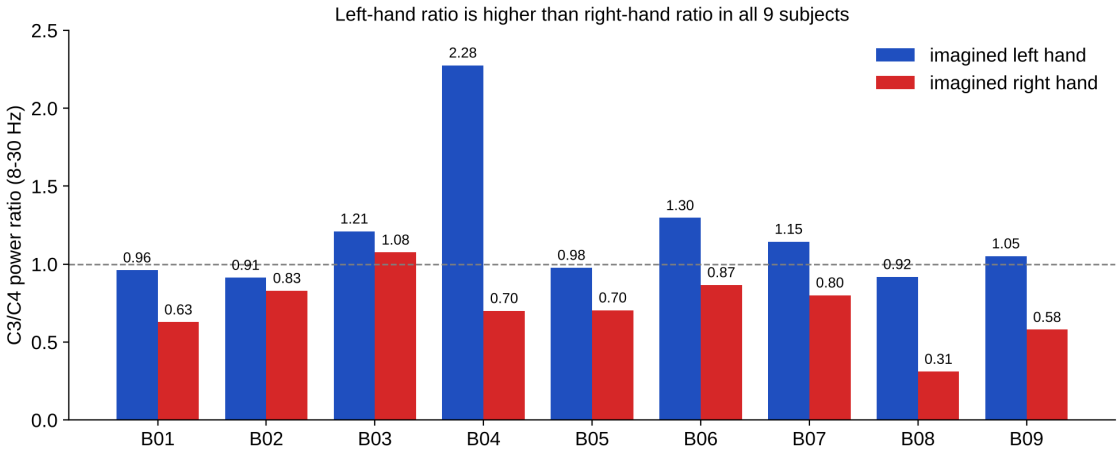


Figure 6.10: C3/C4 power ratio for imagined left and right hand in each BCI Competition IV 2b subject

hand. The figure also shows something important for using the ratio in practice: the value itself changes from person to person. For B03 both values are above 1, and for B08 both are below 1. A fixed rule such as "left if the ratio is above 1" would therefore fail for some people, and the dividing line has to be learned for each person, or the data has to be brought to a common scale first.

To check whether the ratio works on single trials and not only on averages, one number was computed per trial, the logarithm of the C3/C4 power ratio in the 8–30 Hz band from 0.5 to 3.5 s after the cue. A single threshold was learned from each subject's sessions 1 to 3 and then applied, unchanged, to sessions 4 and 5. The same test was done for the other simple single-number features: the power of each channel on its own, the ERD of each channel on its own, and the difference between the ERD of C3 and C4. The results are shown in Figure 6.11.

The log C3/C4 ratio is the best single number by a clear margin. It reaches 73.5 per cent on average on days the threshold never saw, from 55.7 per cent for B02 to 96.6 per cent for B04, and it is better than chance for all nine subjects. When the threshold was learned from eight other people instead of from the subject's own sessions, it still reached 70.8 per cent. The power of either channel on its own reaches only about 63 per cent, and the ERD of either channel on its own about 58 to 60 per cent. There are two reasons why the ratio does so much better. First, anything that makes both channels stronger or weaker together, such as a change in electrode contact, amplifier gain or how alert the

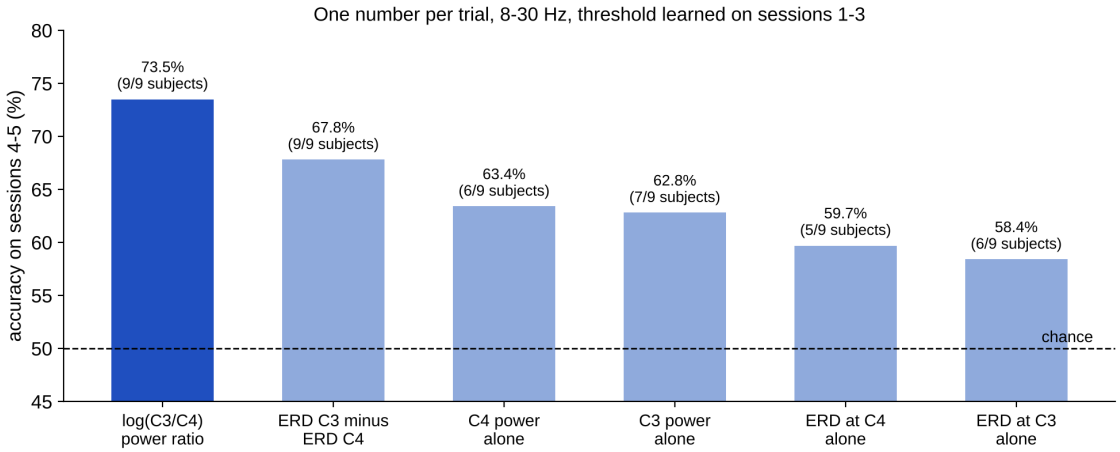


Figure 6.11: Single-trial accuracy of one-number features on sessions 4 and 5 of the BCI Competition IV 2b dataset, mean of nine subjects. The count under each value is the number of subjects for whom the feature was better than chance (binomial test, $p < 0.05$)

person is on that day, cancels out when one channel is divided by the other. Second, the ratio does not need a baseline period at all, while ERD does, and an unreliable baseline was exactly the problem seen in the recorded data. This is the strongest evidence in this work that the C3/C4 ratio is a sound feature for telling left from right, and it supports the decision to include it in the SVM feature sets used on the recorded data.

6.4.5 SVM Training and Classification Accuracy

Several classifiers were trained on the BCI Competition IV 2b data, starting with an SVM on ERD features, which is the closest to what was done for the stroke patients, and then moving to the standard motor imagery methods.

ERD% with an SVM. For each trial, the ERD% at C3, Cz and C4 was computed in the mu band (8–13 Hz) and the beta band (13–30 Hz), giving six features. The rest period was taken from 2 to 1 s before the cue, so that it ends before the warning beep, and the task period from 0.5 to 3.5 s after the cue. Linear and radial basis function SVMs were trained with the MATLAB function `fitcsvm` on standardised features, on sessions 1 to 3, and tested on sessions 4 and 5. Since the ERD curves earlier in this chapter are averages over many trials, the same idea was also tried for the classifier: power was averaged over a group of 5 or 10 trials of the same hand before the ERD was computed. The groups were formed separately inside the training data and inside the test data, so no trial is ever shared between them. The results are given in Table 6.3.

Table 6.3: Accuracy of the ERD% SVM on sessions 4 and 5 of the BCI Competition IV 2b dataset, mean of nine subjects

Trials averaged for one decision	**Linear kernel**	**RBF kernel**
1 (single trial)	65.9%	57.8%
5	77.9%	75.6%
10	80.6%	79.4%

On single trials the linear SVM reaches 65.9 per cent, which is already well above the 50 to 54 per cent obtained when the models were trained on the stroke data and tested on the recorded data (Table 6.6). Averaging trials helps a great deal, raising the accuracy to 80.6 per cent for groups of ten. The price is that one decision then needs ten trials of the same hand, so this is useful for a slow, reliable command but not for fast control. The linear kernel is better than the RBF kernel in every case, which is expected with so few features and so much variation between days. When the same ERD% SVM was trained on eight subjects and tested on the ninth, it reached 62.0 per cent on single trials and 74.4 per cent with ten-trial averaging.

Standard motor imagery classifiers. The covariance-based methods described in Chapter 2 were then trained in Python with scikit-learn [29], all on the 8–30 Hz band and the 0.5 to 3.5 s window after the cue. These were log band power of the three channels with a linear SVM, CSP [9] with LDA and with an SVM, filter bank CSP [7] with an SVM, and a Riemannian tangent space method [11] with an SVM. The SVM settings were chosen by cross-validation inside the training data only. Each model was run with and without Euclidean alignment [31]. Euclidean alignment re-centres the data of each session using the average covariance of that session, so that differences in overall signal size and channel correlation between days or between people are removed before training. It uses no labels, so it can also be applied to the test session without leaking any information. The results for both ways of testing are given in Table 6.4.

The best result when a model is trained on the same person is 80.2 per cent, from filter bank CSP with an SVM and Euclidean alignment. The best result on a person the model has never seen is 74.7 per cent, and every one of the nine subjects is above chance in that test. Three points stand out from the table. Euclidean alignment improved every single model in both tests, by about 3 to 6 percentage points. The choice of model matters very

Table 6.4: Classification accuracy on the BCI Competition IV 2b dataset, mean of nine subjects

Model	**Same person, later days**		**New person**	
	Without EA	**With EA**	**Without EA**	**With EA**
Log band power + SVM	73.0%	78.5%	69.5%	74.7%
CSP + LDA	74.8%	79.1%	70.0%	**74.7%**
CSP + SVM	74.8%	79.0%	70.4%	74.7%
Filter bank CSP + SVM	76.7%	**80.2%**	69.4%	73.5%
Tangent space + SVM	75.5%	78.5%	69.2%	74.6%
log(C3/C4) threshold only	73.5%		70.8%	

little: with alignment, all five models are within about 2 percentage points of each other, and the simplest one, three log band powers with a linear SVM, ties for best on new people. With only three channels, all of these methods end up reading the same small amount of spatial information, so there is little room for a cleverer model to find more. Finally, a single threshold on the C3/C4 ratio, with no SVM at all, is only about 4 to 7 percentage points behind the best models, which again shows how much of the useful information that one number carries.

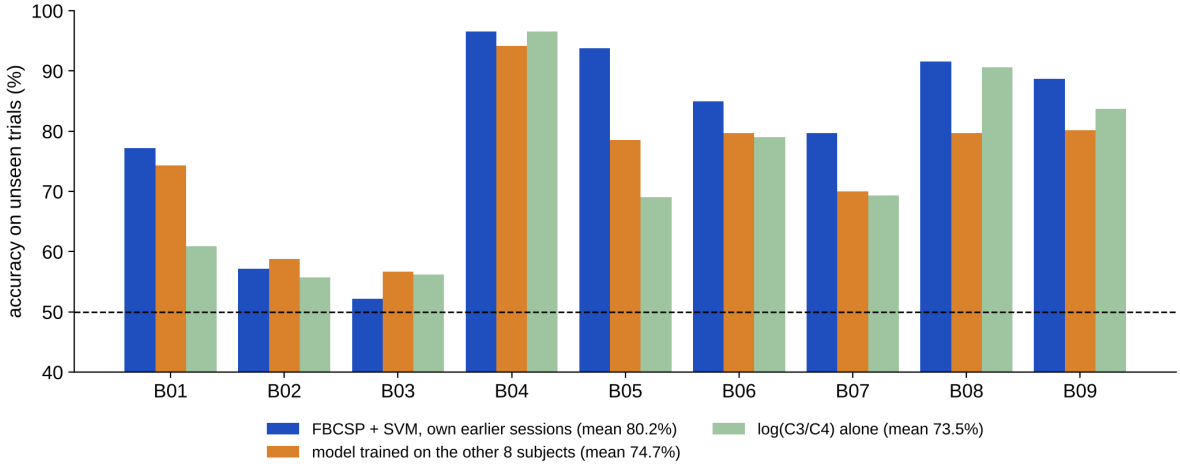


Figure 6.12: Accuracy for each BCI Competition IV 2b subject with the best model for each test and with the C3/C4 ratio alone

Figure 6.12 shows the accuracy for each person. Subject B04 is the best in every test,

with 96.6 per cent when trained on its own earlier sessions, 94.2 per cent from a model that never saw B04, and 96.6 per cent from the C3/C4 ratio alone. B05, B08 and B09 are also close to or above 90 per cent with their own training data. At the other end, B02 and B03 stay between 52 and 59 per cent whatever method is used. This is a well-known effect in BCI research: a fraction of people do not produce a clear enough motor imagery pattern for any current method to pick up, and no change of classifier fixes it.

Table 6.5: Confusion matrix for the new-person test using only C3 and C4, all 6520 trials (CSP + SVM with Euclidean alignment)

	Predicted left	**Predicted right**	**Correct**
True left	2454	806	75.3%
True right	849	2411	74.0%

Table 6.5 gives the confusion matrix for the new-person test using only the two channels the GELO headset has, with every trial of every subject counted once. The overall accuracy is 74.6 per cent, almost the same as with three channels, and the errors are split almost evenly between the two hands. The model is not leaning towards one class, which was a problem with the models trained on the stroke data.

6.4.6 Testing a BCI Competition IV 2b Model on the Recorded Data

Since the 2b model works well on people it has never seen, the next question was whether it works on the data recorded with the GELO headset. A model was trained on all nine 2b subjects using only C3 and C4, with Euclidean alignment, and applied without any change to the 72 usable recorded trials, cut with the same 0.5 to 3.5 s window after each marker. Because each recorded file contains only one class, trials from the same file are not independent, so the fair count is per file, giving 15 decisions.

The result was at chance. Across eight combinations of model and alignment, between 4 and 6 of the 15 files were classified correctly, and the best permutation test gave $p = 0.83$, meaning a result this good is expected most of the time even with random labels. The predictions were split roughly evenly between left and right, so the failure is not a bias towards one class. To see why, the separation given by the log C3/C4 ratio was measured directly on both datasets, using the area under the ROC curve, where 0.5 means no

separation and values further from 0.5 mean better separation. For the 2b data it was 0.22, a distance of 0.28 from 0.5. For the 72 recorded trials it was 0.51, a distance of only 0.01. Taken over all recorded trials, the left and right difference in the C3/C4 balance is therefore about twenty times weaker than in 2b, even though single trials such as the one in Figure 6.14 show a large difference.

This is a useful negative result. The same code and the same model reach 74.6 per cent on new people in 2b, so the classifier itself is working. What stands between it and the GELO headset is the recorded data, most likely the way it was collected: each class was recorded in separate files, markers were placed by hand, the baseline is not a true rest period, and there are only 72 trials. The BCI Competition IV 2b protocol, with left and right trials mixed within one session, fixed cue timing and several sessions per person, is a direct model for how the next GELO recordings should be made.

6.5 Results on the Recorded Data

6.5.1 ERD/ERS of the Recorded Data

The same analysis was then applied to the data recorded with the GELO headset. Figure 6.13 shows the result for one subject over the four second imagination window.

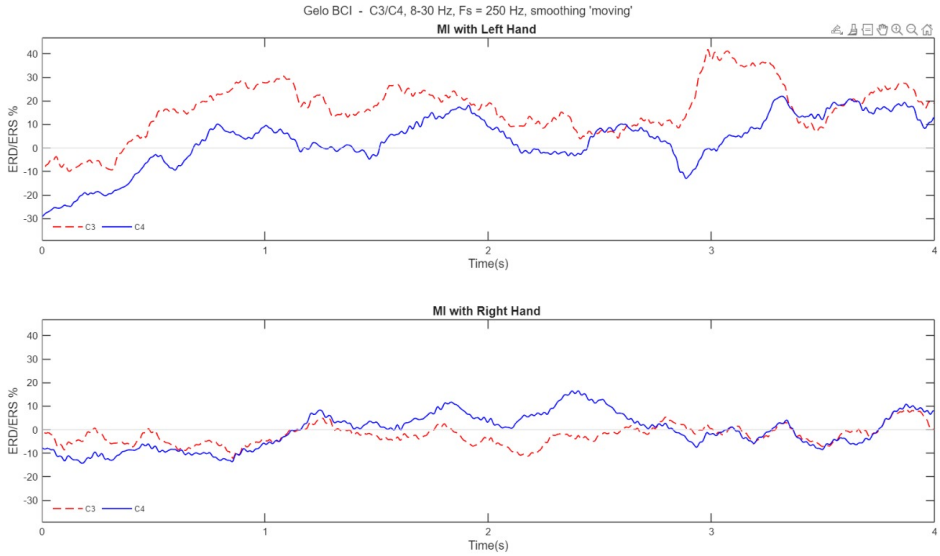


Figure 6.13: ERD/ERS of C3 and C4 for the data recorded in this project, 8–30 Hz, $F_s = 250$ Hz

Two things can be seen from this figure. The first is that the two classes do separate. For

left-hand imagery the C3 trace stays above C4 for most of the window, reaching 20 to 40 per cent, while for right-hand imagery the order reverses and C4 stays above C3, with both traces remaining within roughly ±15 per cent. The sign of the difference between the two channels therefore flips between the two classes, which is the property a classifier can make use of.

The second is that the separation does not take the same form as in the reference dataset. In the reference recording the contralateral channel falls below the baseline, giving a clear desynchronisation, whereas in the recorded data both channels mostly rise above the baseline and the difference between them is what carries the class information. There are two likely reasons for this. The baseline period in the recorded sessions is taken from the seconds immediately before the marker, during which the subject is already preparing for the task, so the reference power is not a true rest level. In addition, the four-channel dry headset picks up more muscle and movement artefact than the 30-channel saline system used for the reference dataset, and these artefacts raise the power in the 8–30 Hz band. The result is still usable, but it means the absolute ERD values from this setup should not be compared directly against published values.

6.5.2 Power Spectral Density for Left and Right Imagery

The PSD of the two motor cortex channels was computed for each trial. Figure 6.14 shows a left-hand trial and Figure 6.15 a right-hand trial from the same subject. In both figures the upper two plots show the two channels separately and the lower plot overlays them, with the spectral peak marked.

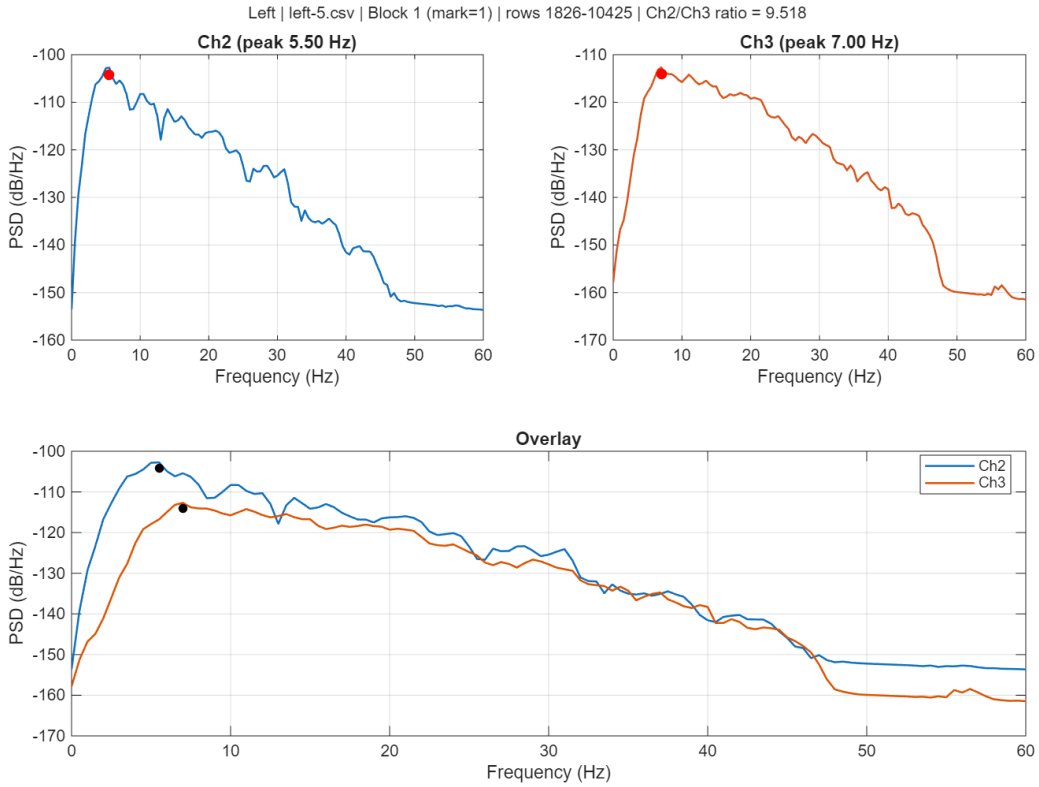


Figure 6.14: Power spectral density of C3 and C4 for an imagined left-hand grasp

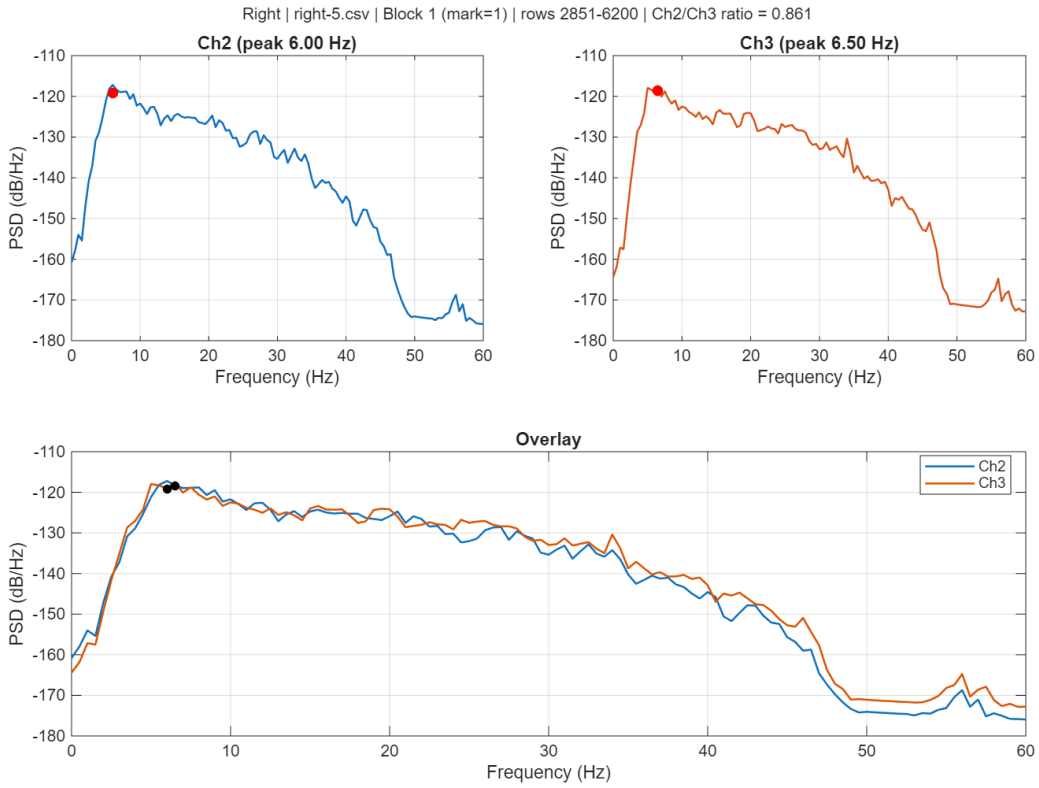


Figure 6.15: Power spectral density of C3 and C4 for an imagined right-hand grasp

The difference between the two cases is clearest in the overlay plots. For the left-hand trial the C3 curve sits roughly 10 dB above the C4 curve across most of the low frequency range, giving a C3 to C4 power ratio of 9.52. For the right-hand trial the two curves lie almost on top of each other, with a ratio of 0.86. The spectral peaks are close together in both cases, at 5.50 Hz and 7.00 Hz for the left trial and 6.00 Hz and 6.50 Hz for the right trial, which shows that the peak frequency itself does not separate the classes. It is the balance of power between the two hemispheres, and not the shape of the spectrum, that carries the useful information. This is the reason the logarithm of the C3 to C4 power ratio was included as a feature for the classifier.

Both figures also show that the signal falls away sharply beyond about 45 Hz. This is the effect of the filtering applied inside the device, and it is why the analysis in this work is restricted to the 8–30 Hz range for motor imagery.

6.5.3 Classification Results

Support Vector Machine classifiers were trained using the reference dataset [4] as training data and the data recorded in this project as test data. Three feature sets were tried, each with both a linear and a radial basis function kernel, and each using absolute power as well as ERD power:

- **SVM-A:** mean, median, skewness and kurtosis of C3 and C4 (8 features)
- **SVM-B:** the above, plus $\log_{10}(P_{C3}/P_{C4})$ and the Welch band power over 8–30 Hz (9 features)
- **SVM-C:** the C3 to C4 power ratio alone (1 feature)

The accuracies obtained are summarised in Table 6.6.

All six combinations give accuracies between 50 and 54 per cent, which is close to what would be obtained by guessing on a two-class problem. Adding the power ratio to the statistical features in SVM-B did not improve the result, and using the ratio on its own in SVM-C performed worst. Using ERD power instead of absolute power made very little difference either way.

This result should be read together with the benchmarks reported for the reference dataset itself, where CSP with LDA reached 55.6 per cent, filter bank CSP with an SVM reached 57.6 per cent and the best reported method reached 72.21 per cent, all using 30 channels

Table 6.6: Classification accuracy of the trained SVM models on the recorded data

Model	**Features**	**Absolute power**	**ERD power**
SVM-A	Mean, median, skewness, kurtosis of C3 and C4	53.9%	53.1%
SVM-B	SVM-A features + C3/C4 PSD ratio	53.7%	53.5%
SVM-C	C3/C4 PSD ratio only	50.2%	51.4%

on data from the same recording system [4]. Against that background, the figures in Table 6.6 are not surprising. Two difficulties combine here. The first is that the model is trained on one recording system and tested on another, so the feature distributions do not match: the training data comes from stroke patients recorded with 30 saline electrodes at 500 Hz, while the test data comes from healthy subjects recorded with four dry electrodes at 250 Hz. The second is that only two channels are available for the spatial information, so spatial filtering methods such as CSP [9], which are what give the published methods most of their accuracy, cannot be applied at all. The earlier ERD and PSD results show that a difference between the classes does exist in the recorded data, so the limitation lies in the training strategy and the number of channels rather than in the data being empty of information.

The BCI Competition IV 2b results in Section 6.4 put these numbers in perspective. On that dataset, a model using only C3 and C4 and trained on other people reached 74.6 per cent on a new person, so two channels are enough when the recordings are clean and well timed. The same section also shows that this working model still gives chance-level results on the recorded trials. Together these point to the training strategy and, above all, the way the GELO data was recorded as the main things to improve, rather than the number of channels.

6.6 Cognitive Assessment Results

The second part of the system computes a cognitive score from the same recording. Results for this part are reported from a complete monitoring session.

6.6.1 Signal Quality and Filtering

Since the whole cognitive score depends on band power, the quality of the contact between the electrodes and the skin matters a great deal. Figure 6.16 shows how the resistance of the four electrodes changed over the first six minutes after the headset was put on.

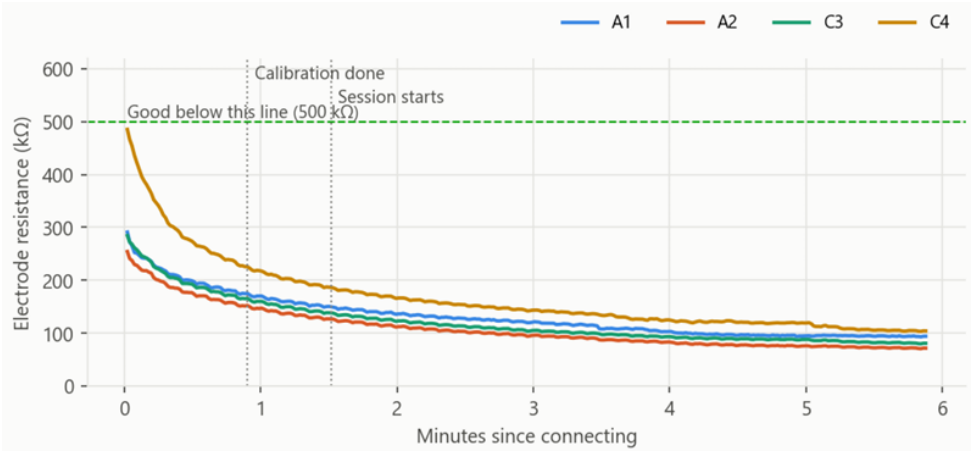


Figure 6.16: Electrode resistance of the four channels after connecting the headset

All four electrodes start between 250 and 500 kΩ and settle to between 70 and 110 kΩ within about four minutes, which is well inside the usable range. C4 settles highest and takes longest, which is consistent with it being the channel that gave the noisiest motor imagery traces. The figure also shows why the session is not started immediately: calibration is performed at around one minute and the session begins at about one and a half minutes, by which time the contact has largely stabilised.

Two filters are applied to the continuous signal before the band powers are computed. Their responses are shown in Figure 6.17. A 0.5 Hz high-pass filter removes the slow drift caused by sweat and small movements of the electrodes while leaving the delta band intact, and a 50 Hz notch filter removes mains interference. Both filters process the signal block by block while keeping their internal state, so that no discontinuity is introduced at the block boundaries.

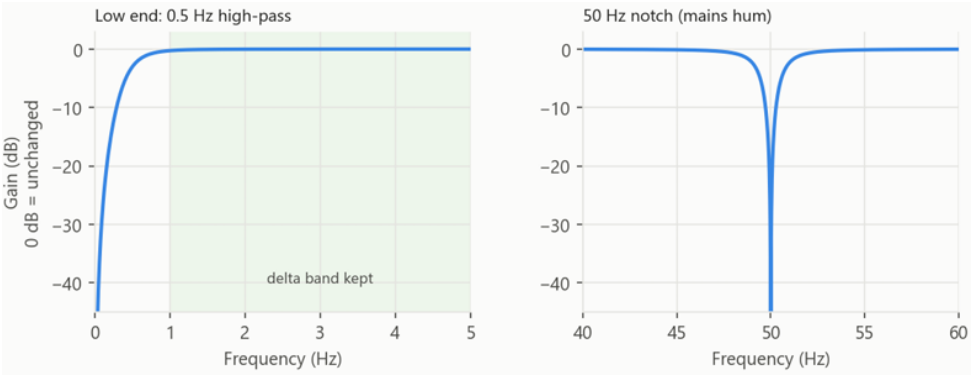


Figure 6.17: Frequency response of the 0.5 Hz high-pass and 50 Hz notch filters

6.6.2 Usable Frequency Range

Figure 6.18 shows how much of the original amplitude survives at each frequency once the device filtering and the filters above are taken together.

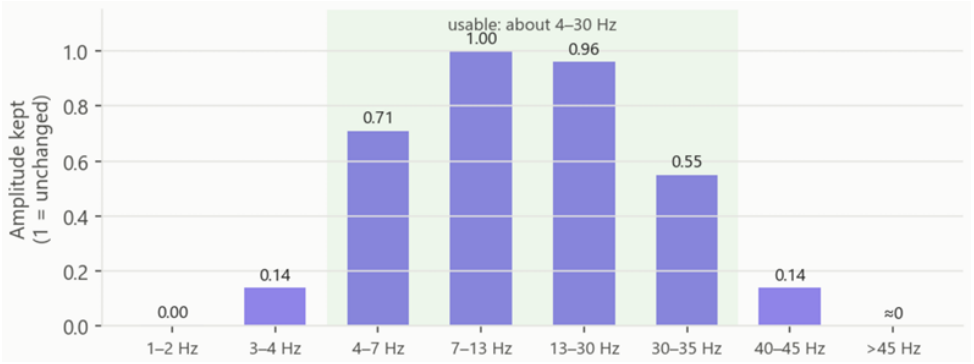


Figure 6.18: Fraction of the original amplitude retained across frequency bands

The 7–13 Hz and 13–30 Hz ranges pass almost unchanged, at 1.00 and 0.96, and the 4–7 Hz range is retained at 0.71. Below 4 Hz the response falls away sharply, to 0.14 in the 3–4 Hz range and effectively to zero below 2 Hz, and above 35 Hz it falls away in the same way. The practical consequence is that the delta band cannot be measured reliably with this device and the gamma band is only partly available, so the effective measurement range for cognitive scoring is about 4 to 30 Hz. The scores used in this work are therefore built from theta, alpha and beta, which all lie inside that range.

6.6.3 Stress and Fatigue Reference Levels

Stress is estimated from the arousal ratio, which is the relative high-beta power in the 20–30 Hz range divided by the relative alpha power [25]. When a person becomes tense,

beta activity rises and alpha activity falls, so this ratio increases. Figure 6.19 shows how the ratio behaved over one session.

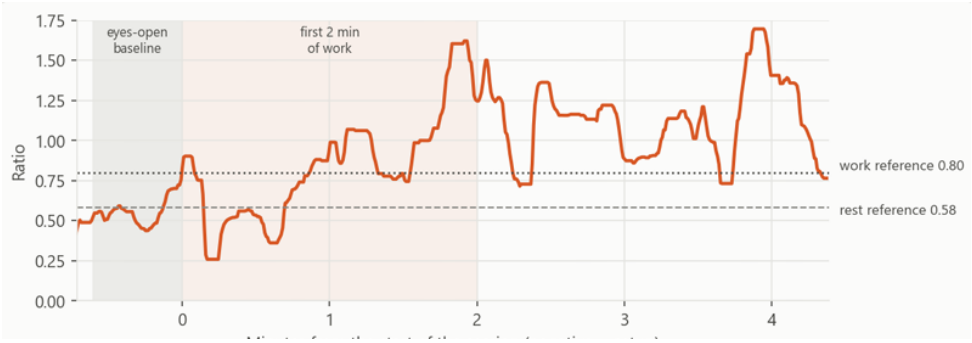


Figure 6.19: Arousal ratio over a session, with the rest and work reference levels marked

During the eyes-open baseline the ratio stays near 0.5, and a rest reference of 0.58 is established from this period. Once work begins the ratio rises and moves between about 0.75 and 1.70 for the rest of the session. A work reference of 0.80 is taken from the first two minutes of work, which are deliberately excluded from the stress calculation so that the settling-in period is not reported as stress. The score itself is obtained by comparing the current ratio against this reference,

$$\text{Score} = 50 + 50 \tanh\left(\log_2 \frac{\text{ratio}}{\text{reference}}\right) \tag{6.2}$$

which maps the ratio onto a 0 to 100 scale, with 50 meaning the person is at their own reference level. The displayed value is the median over the last 30 seconds, so that a single noisy block does not move the score. Fatigue is computed in the same way from the ratio of combined theta and alpha power to beta power.

6.6.4 Complete Session Output

Figure 6.20 shows the interface at the end of a monitoring session, with all the outputs computed from the single four-channel recording.

At the instant captured, attention is 91 and cognitive load is 87, both reported as high, while relaxation is 23. Stress and fatigue are shown as not yet available, since both require the opening period of the session to pass before a reference can be established. The session plot in the middle shows how attention and cognitive load changed over the minute of recording. The two do not move together throughout: around the half minute mark cognitive load drops to near 10 while attention stays above 40, and both then climb



Figure 6.20: Cognitive score interface showing the session output

back together towards the end. This is useful because it shows the two measures are not simply reporting the same quantity under different names.

The brain wave panel on the right shows the change in each band relative to the resting baseline. Delta is down 92 per cent, theta down 83 per cent, alpha down 69 per cent, beta down 50 per cent and gamma down 6 per cent. The pattern is consistent with an alert, engaged state, since the slow bands associated with drowsiness are suppressed much more than the faster bands associated with active thinking. The delta figure should not be read too closely, as Figure 6.18 shows that very little of the delta band survives the filtering on this device.

The sidebar also shows that the session follows a fixed sequence of setup, connection, signal check, calibration, baseline and then the session itself, and the signal quality indicator stayed green throughout, confirming that the contact held for the full recording.

6.7 Reason for the Observed Variations

The differences seen between the two motor imagery classes come from the way the motor cortex is organised. Imagining a movement of one hand changes the activity over the opposite hemisphere, so the balance of power between C3 and C4 shifts depending on which hand is imagined. This is what both the ERD traces in Figure 6.13 and the power

ratios in Figures 6.14 and 6.15 show, and it is the only property of the signal that separated the two classes reliably in this work.

The difference between the recorded data and the reference dataset has two main causes. The headset used here has four dry electrodes while the reference system had thirty saline electrodes, so both the signal quality and the amount of spatial information available are lower. The baseline used for the ERD calculation is also taken from the moments just before the marker, when the subject is already preparing for the task, so it does not represent a true rest level and the computed percentages are shifted upward as a result.

The low classification accuracy follows from the same two causes, together with the decision to train on one dataset and test on another. The subjects differ, the recording hardware differs, the sampling rate differs and the number of channels differs, so features computed on the training set do not fall in the same range as those computed on the test set. The small amount of recorded data, 50 right-hand and 22 left-hand usable trials, also means the two classes are unbalanced, which pushes a classifier towards the majority class and keeps the accuracy near the chance level.

For the cognitive scoring, the variation between sessions is handled by referencing everything to the person's own calibration. Because the scores are computed from relative band powers and compared against a baseline recorded at the start of each session, a change in overall signal level caused by electrode contact improving or worsening does not by itself change the score.

The BCI Competition IV 2b data shows a different kind of variation, between people rather than between devices. All nine subjects show the expected change in the C3/C4 balance, but its size differs a lot, from a ratio of 2.28 against 0.70 for B04 to 0.91 against 0.83 for B02. The overall level of the ratio also differs, so that for some people both values sit above 1 and for others both sit below 1. This is most likely due to the exact position of the electrodes over each person's motor cortex, the shape of the head and how strongly each person can imagine the movement, and it is the reason the decision threshold has to be learned for each person or the data aligned first.

6.8 Summary of Analysis

The BCI Competition IV 2b dataset confirms that the analysis method and the classifiers used in this project work. With the same ERD/ERS recipe as for the stroke patients,

the channel opposite to the imagined hand drops more in 16 of 18 cases, and subject B04 shows a textbook pattern, with C4 falling to about −50 per cent for left-hand imagery and C3 to about −60 per cent for right-hand imagery. The C3/C4 power ratio is higher for the left hand than for the right hand in all nine subjects, and as a single number it identifies the hand on unseen days with 73.5 per cent accuracy, better than chance for every subject and 96.6 per cent for B04. The best SVM reached 80.2 per cent when trained on the same person and 74.7 per cent on a new person, and the model choice mattered far less than Euclidean alignment or the quality of the data.

The recorded GELO data shows the same kind of difference in the examples analysed, with the C3 to C4 power ratio reaching 9.52 for a left-hand trial and 0.86 for a right-hand trial on the same subject, and the direction of the difference between the two channels reversing between the classes. Taken over all 72 recorded trials, however, the difference is much weaker than in the 2b data.

The classifiers trained on the stroke data and tested on the recorded data do not convert this difference into useful accuracy. All six combinations of feature set and kernel gave between 50 and 54 per cent on a two-class problem. Since the published benchmarks on the reference dataset reach only 55 to 72 per cent using thirty channels and matched training and test conditions, the result is in line with the difficulty of the problem, but it confirms that training on a public dataset and testing on data from a different device is not a workable strategy. The same conclusion holds for a model trained on the BCI Competition IV 2b data, which works well on new 2b subjects but not on the recorded trials. The next step is to record a larger and better balanced set of trials on the headset itself and to train and test within that data, rather than across the two sources.

The cognitive assessment part of the system is working end to end. Electrode impedance settles into the usable range within a few minutes, the filtering gives a reliable measurement band of roughly 4 to 30 Hz, and attention, cognitive load, relaxation, stress and fatigue are computed from the band powers and displayed live against the person's own calibrated reference. The session output in Figure 6.20 shows attention and cognitive load varying independently over a recording, which indicates that the scores are responding to the signal rather than tracking a single underlying quantity.

6.9 Work carried out so far

The work completed so far covers the following stages. A review of existing research was carried out on EEG signal processing, motor imagery decoding, frequency band analysis and machine learning methods for EEG, and the approach for the project was decided on the basis of that review. The hardware was then set up using the GELO BCI headset, and a recording procedure was established covering electrode impedance checking, alpha peak calibration, baseline calibration and marker placement.

The ERD/ERS analysis reported in the reference dataset was reproduced to confirm that the processing chain was correct, after which EEG data was recorded for left-hand and right-hand motor imagery and the same analysis applied to it. The channels involved in left and right motor imagery were identified, and the recorded data was compared against the published results. The same analysis was then carried out on all nine subjects of the BCI Competition IV 2b dataset, covering ERD/ERS, PSD, the C3/C4 power ratio and a set of SVM and CSP classifiers tested on later sessions and on new people, and a 2b model was tested on the recorded data. Several SVM models were then trained using different feature sets and kernels, and the features that separate the two classes were identified. In parallel, the cognitive scoring system was built and tested over complete sessions, covering calibration, band power computation, score calculation and the display interface.

Chapter 7

Conclusion and Future Work

7.1 Conclusion

This project set out to build an EEG system around a four-channel dry-electrode headset that can tell left-hand from right-hand motor imagery and, from the same recording, report the mental state of the person as a cognitive score.

The motor imagery analysis was built and checked in three stages. First, the ERD/ERS analysis of the published stroke patient dataset was reproduced, which confirmed that the processing chain gives the same result as the original paper. Second, the same method was applied to all nine healthy subjects of the BCI Competition IV 2b dataset, which matches the headset in sampling rate and electrode positions. Here the expected pattern appeared clearly: the 8–30 Hz power dropped more over the side of the head opposite to the imagined hand in 16 of 18 cases, and subject B04 showed the textbook result, with C4 falling to about -50 per cent during left-hand imagery and C3 to about -60 per cent during right-hand imagery. Third, the analysis was applied to left and right motor imagery recorded with the GELO headset.

The most useful single finding is the C3/C4 power ratio. On the 2b data it was higher for the left hand than for the right hand in all nine subjects, and as a single number per trial it identified the hand on unseen days with 73.5 per cent accuracy, better than chance for every subject and 96.6 per cent for B04. It works better than the power or the ERD of either channel on its own because it cancels changes that affect both channels together and needs no rest baseline. The best classifiers reached 80.2 per cent when trained on the same person and 74.7 per cent on a person never seen in training, using filter bank CSP or simple log band power with an SVM and Euclidean alignment. With only two or three channels the choice of classifier made very little difference, while Euclidean alignment helped every model.

On the data recorded with the GELO headset, the same kind of C3/C4 difference is visible in the trials examined, with a power ratio of 9.52 for a left-hand trial and 0.86 for a right-

hand trial. However, classifiers trained on either public dataset stayed close to chance on the recorded data, and across all 72 usable trials the left and right difference was much weaker than in the 2b data. Since the same classifier works well on new people in 2b, the main limit is the recorded data and the way it was collected, and not the classifier or the number of channels.

The cognitive assessment part works end to end. The electrode contact settles within a few minutes, the usable measurement band of the headset was found to be about 4 to 30 Hz, and attention, cognitive load, relaxation, stress and fatigue are computed live and compared against each person's own calibration, so that changes in overall signal level do not move the scores.

7.2 Future Work

The results point clearly to what should be done next, in order of expected benefit.

- **A better recording protocol.** New GELO recordings should follow the BCI Competition IV 2b design: left and right trials mixed randomly within one session, cues and markers placed by software at fixed times, a calm fixation period before each cue to act as a true baseline, equal numbers of trials for each hand, and several sessions on different days.

- **Per-user calibration.** Training on the same person's earlier sessions was worth about 5 percentage points over a model trained on other people, and the C3/C4 threshold differs between people. A short calibration run of cued trials at the start of use is a cheap way to gain this.

- **Testing the 2b model again on new GELO data.** Once data recorded with the new protocol is available, the 2b model with Euclidean alignment can be applied to it directly, with the 74.7 per cent reached on new 2b subjects as the reference.

- **Real-time operation on the Raspberry Pi 5.** The C3/C4 ratio and the log band power SVM are very light to compute, so both the motor imagery decision and the cognitive score can run live on the embedded platform described in Chapter 4.

- **Deep learning and imagined speech.** Models such as convolutional transformers need far more data than is available now. With a larger and cleaner dataset recorded

on the headset, they become worth trying, both for motor imagery and for the longer-term aim of decoding imagined words.

Bibliography

[1] World Health Organization, "Over 1 in 3 people affected by neurological conditions, the leading cause of illness and disability worldwide," *WHO*, Mar. 14, 2024. [Online]. Available: `https://www.who.int/news/item/14-03-2024-over-1-in-3-people-affected-by-neurological-conditions--the-leading-cause-of-illness-and-disability-worldwide`. [Accessed: April 10, 2026].

[2] T. R. Mullen, C. A. E. Kothe, Y. M. Chi, A. Ojeda, T. Kerth, S. Makeig, T.-P. Jung, and G. Cauwenberghs, "Real-time neuroimaging and cognitive monitoring using wearable dry EEG," *IEEE Transactions on Biomedical Engineering*, vol. 62, no. 11, pp. 2553–2567, 2015.

[3] S. L. Kappel, M. L. Rank, H. O. Toft, M. Andersen, and P. Kidmose, "Dry-contact electrode ear-EEG," *IEEE Transactions on Biomedical Engineering*, vol. 66, no. 1, pp. 150–158, Jan. 2019.

[4] H. Liu, P. Wei, H. Wang, X. Lv, W. Duan, M. Li, Y. Zhao, Q. Wang, X. Chen, G. Shi, B. Han, and J. Hao, "An EEG motor imagery dataset for brain computer interface in acute stroke patients," *Scientific Data*, vol. 11, no. 1, p. 131, 2024, doi: 10.1038/s41597-023-02787-8.

[5] R. Leeb, F. Lee, C. Keinrath, R. Scherer, H. Bischof, and G. Pfurtscheller, "Brain-computer communication: motivation, aim, and impact of exploring a virtual apartment," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 15, no. 4, pp. 473–482, 2007.

[6] M. Tangermann, K.-R. Müller, A. Aertsen, N. Birbaumer, C. Braun, C. Brunner, R. Leeb, C. Mehring, K. J. Miller, G. R. Müller-Putz, G. Nolte, G. Pfurtscheller, H. Preissl, G. Schalk, A. Schlögl, C. Vidaurre, S. Waldert, and B. Blankertz, "Review of the BCI competition IV," *Frontiers in Neuroscience*, vol. 6, art. 55, 2012, doi: 10.3389/fnins.2012.00055.

[7] K. K. Ang, Z. Y. Chin, C. Wang, C. Guan, and H. Zhang, "Filter bank common spatial pattern algorithm on BCI competition IV datasets 2a and 2b," *Frontiers in Neuroscience*, vol. 6, art. 39, 2012, doi: 10.3389/fnins.2012.00039.

[8] G. Pfurtscheller and F. H. Lopes da Silva, "Event-related EEG/MEG synchronization and desynchronization: basic principles," *Clinical Neurophysiology*, vol. 110, no. 11, pp. 1842–1857, 1999.

[9] H. Ramoser, J. Müller-Gerking, and G. Pfurtscheller, "Optimal spatial filtering of single trial EEG during imagined hand movement," *IEEE Transactions on Rehabilitation Engineering*, vol. 8, no. 4, pp. 441–446, 2000.

[10] C. Cortes and V. Vapnik, "Support-vector networks," *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995.

[11] A. Barachant, S. Bonnet, M. Congedo, and C. Jutten, "Multiclass brain-computer interface classification by Riemannian geometry," *IEEE Transactions on Biomedical Engineering*, vol. 59, no. 4, pp. 920–928, 2012.

[12] C. Xu and R.-Z. Xia, "EEG signal classification and feature extraction methods based on deep learning: A review," in *Proc. 2023 2nd International Conference on Big Data, Information and Computer Network (BDICN)*, Xishuangbanna, China, 2023, pp. 203–206, doi: 10.1109/BDICN58493.2023.00046.

[13] Y. Song, Q. Zheng, B. Liu, and X. Gao, "EEG Conformer: Convolutional transformer for EEG decoding and visualization," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 31, pp. 710–719, 2023, doi: 10.1109/TNSRE.2022.3230250.

[14] J. T. Panachakel and A. G. Ramakrishnan, "Decoding imagined speech from EEG using transfer learning," *IEEE Access*, vol. 9, pp. 135371–135383, 2021, doi: 10.1109/ACCESS.2021.3116196.

[15] J. Su, J. Wang, W. Wang, Y. Wang, C. Bunterngchit, P. Zhang, and Z.-G. Hou, "An adaptive hybrid brain-computer interface for hand function rehabilitation of stroke patients," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 32, pp. 2950–2960, 2024.

[16] G. Krishna, C. Tran, J. Yu, and A. H. Tewfik, "Speech recognition with no speech or with noisy speech," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 1090–1094, 2019.

[17] G. Krishna, C. Tran, Y. Han, M. Carnahan, and A. H. Tewfik, "Speech synthesis using EEG," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 1235–1239, 2020.

[18] S. Zhao and F. Rudzicz, "Classifying phonological categories in imagined and articulated speech," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Brisbane, Australia, Apr. 2015, pp. 992–996.

[19] L. Bollens, C. Piffay, B. Accou, J. Vanthornhout, H. Van Hamme, and T. Francart, "Auditory EEG decoding challenge for ICASSP 2024," *IEEE Open Journal of Signal Processing*, vol. 6, pp. 1–13, 2025, doi: 10.1109/OJSP.2025.3534122.

[20] J. J. Bird, A. Ekárt, L. J. Manso, D. R. Faria, and E. P. Ribeiro, "A study on mental state classification using EEG-based brain-machine interface," in *Proc. 2018 International Conference on Intelligent Systems (IS)*, pp. 1–6, 2018.

[21] S. Jawed, I. Faye, and A. S. Malik, "Deep learning-based assessment model for real-time identification of visual learners using raw EEG," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 32, pp. 378–390, 2024, doi: 10.1109/TNSRE.2024.3351694.

[22] Y. Ming, D. Wu, Y.-K. Wang, Y. Shi, and C.-T. Lin, "EEG-based drowsiness estimation for driving safety using deep Q-learning," *IEEE Transactions on Emerging Topics in Computational Intelligence*, vol. 5, no. 4, pp. 583–594, 2021, doi: 10.1109/TETCI.2020.2997031.

[23] E. Jiri, A. Maraoui, I. Werda, "Driver drowsiness detection using boosting algorithms with integrated EEG spectral entropy and PSD features," in *Proc. 2025 IEEE 22nd International Conference on Sciences and Techniques of Automatic Control and Computer Engineering (STA)*, pp. 253–257, 2025.

[24] H. H. Jasper, "The ten-twenty electrode system of the International Federation," *Electroencephalography and Clinical Neurophysiology*, vol. 10, pp. 371–375, 1958.

[25] GELO, "GELO BCI Digital EEG Recorder Headphones: user manual and SDK documentation," GELO, 2026. [Online]. Available: https://www.gelo.one/. [Accessed: April 10, 2026].

[26] Raspberry Pi Ltd., "Raspberry Pi 5 product brief," 2023. [Online]. Available: https://datasheets.raspberrypi.com/rpi5/raspberry-pi-5-product-brief.pdf. [Accessed: April 10, 2026].

[27] Waveshare, "7inch HDMI LCD capacitive touch display," Waveshare Wiki. [Online]. Available: https://www.waveshare.com/wiki/7inch_HDMI_LCD. [Accessed: April 10, 2026].

[28] A. Gramfort, M. Luessi, E. Larson, D. A. Engemann, D. Strohmeier, C. Brodbeck, R. Goj, M. Jas, T. Brooks, L. Parkkonen, and M. S. Hämäläinen, "MEG and EEG data analysis with MNE-Python," *Frontiers in Neuroscience*, vol. 7, art. 267, 2013.

[29] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[30] P. Welch, "The use of fast Fourier transform for the estimation of power spectra: A method based on time averaging over short, modified periodograms," *IEEE Transactions on Audio and Electroacoustics*, vol. 15, no. 2, pp. 70–73, 1967.

[31] H. He and D. Wu, "Transfer learning for brain-computer interfaces: A Euclidean space data alignment approach," *IEEE Transactions on Biomedical Engineering*, vol. 67, no. 2, pp. 399–410, 2020.

Appendices

Appendix A

Program Outcomes and Program Specific Outcomes addressed

#	POs and PSOs	Contribution from the Project	Level
1	Engineering Knowledge:	The project applies knowledge of mathematics, signal processing, and machine learning to process and classify EEG signals. Concepts such as frequency transforms, filtering, and statistical analysis are used throughout the system.	3
2	Problem Analysis:	The problem of detecting cognitive states from EEG signals is analyzed by reviewing existing research and identifying key challenges such as signal noise, artifact removal, and variability across subjects.	3
3	Design/development of solutions	A complete EEG-based cognitive assessment system will be designed and developed, covering signal acquisition, feature extraction from frequency bands, and classification using a machine learning model.	3
4	Conduct investigations of complex problems:	EEG data will be collected from subjects performing specific cognitive tasks. The data will be analyzed and used to train and evaluate the classification model.	2

#	POs and PSOs	Contribution from the Project	Level
5	Engineering tool usage:	Tools such as Python, MATLAB, and EEG acquisition hardware are used to build and test the system. Libraries like scikit-learn and MNE-Python will be used for signal processing and machine learning.	2
6	The Engineer and the world:	The system has practical applications in healthcare and education, such as monitoring mental workload and detecting early signs of cognitive decline.	2
7	Ethics:	Since the project involves collecting brain signals from human subjects, informed consent is obtained before data collection. Data privacy and responsible use of personal biological data is considered.	2
8	Individual and Team Work:	The project is carried out as a team where different members handled hardware setup, signal processing, and model development. Each member also contributed independently to their assigned tasks.	2
9	Communication:	The project work is documented in the form of a technical report and presented to faculty through regular reviews and a final presentation. Results and methodology were clearly explained.	2

#	POs and PSOs	Contribution from the Project	Level
10	Project Management and Finance:	The project is planned with defined objectives covering data collection, model development, and testing phases. Resources are managed within the available budget and time.	1
11	Life-long Learning:	Working on this project requires learning topics beyond the regular syllabus, such as brain-computer interfaces and deep learning for time-series data, encouraging independent and continuous learning.	2
1	PSO1:	The project directly involves signal processing and embedded system concepts. EEG signals are acquired, filtered, and processed to extract frequency band features, which are then used to classify cognitive states.	3
2	PSO2:	The problems of motor imagery classification and cognitive state identification are solved using modern machine learning algorithms and tools, demonstrating practical problem-solving skills in the area of embedded and communication systems.	3

PSO1:The ability to analyze and design systems in the areas related to microelectronics, Communication, Signal Processing and embedded systems for solving real world problems (Professional Skills).

PSO2:The ability to identify problems in the areas of communication and embedded systems and provide efficient solutions using modern tools/algorithm individually or working in a team (Problem solving Skills)

Levels: Poor:1, Good :2, Excellent:3

Appendix B

Sustainable Development Goals addressed

Level 3 :Strong Alignment (Direct and significant contribution to the SDG).

Level 2 :Moderate Alignment (Indirect or supporting contribution to the SDG).

Level 1 :Basic Alignment (Minimal or enabling contribution).

SDGs:

1. End poverty in all its forms everywhere

2. End hunger, achieve food security and improved nutrition and promote sustainable agriculture

3. Ensure healthy lives and promote well-being for all at all ages

4. Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all

5. Achieve gender equality and empower all women and girls

6. Ensure availability and sustainable management of water and sanitation for all

7. Ensure access to affordable, reliable, sustainable and modern energy for all

8. Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all

9. Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation

10. Reduce inequality within and among countries

11. Make cities and human settlements inclusive, safe, resilient and sustainable

12. Ensure sustainable consumption and production patterns

13. Take urgent action to combat climate change and its impacts

14. Conserve and sustainably use the oceans, seas and marine resources for sustainable development

15. Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss

16. Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels
17. Strengthen the means of implementation and revitalize the Global Partnership for Sustainable Development

SDG No. and Name	**SDG Type**	**SDG Target**	**Mapping Level**	**Justification**	**Impact Domain**
SDG 3: Good Health and Well-being	Secondary	3.d	Level 2 (Moderate)	The system uses EEG signals to monitor brain activity and detect stress levels. It also support early identification of mental stress of a person.	Social
SDG 4: Quality Education	Primary	4.a	Level 3 (Strong)	The project can be used to monitor attention and stress levels during learning. It may also help disabled people communicate using imagined speech.	Social
SDG 8:Decent Work and Economic Growth	Secondary	8.5	Level 2 (Moderate)	The project helps the disabled people communicate using imagined speech systems and support them in workplace.	Social